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A Physics-Informed Neural Network Approach for Invasive Mechanical Ventilation Weaning Prediction in ICU Patients

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Chapter 1

Abstract

The premature extubation of an invasively ventilated patient can result in reintubation, further respiratory support, as well as exposure to significantly higher risk of physical complications. On the other hand, unnecessary prolonged invasive ventilation could also lead to increased risk of physical complications, such as severe pulmonary infections. Accurately identifying when a patient is ready to be extubated remains a critical and challenging decision within Intensive Care Units (ICUs). This project explores whether time-series neural networks could support patient extubation decisions, and whether clinically grounded physiological constraints could further improve the consistency and explainability of the predictions.

An extubation episode cohort was constructed from the MIMIC-IV v3.1 dataset, consisting of a total of 7,706 extubation episodes based on 7,085 unique patients. The extubation failure label was defined as satisfying either invasive reintubation within 48 hours past the extubation event, or non invasive repiratory support for longer than 48 hours past the extubation event. The extubation failure class accounts for 784 extubation episodes of the cohort. The feature space consists of 17 dynamic physiological features and 3 static features with values extracted within a 6 hour decision window prior to the extubation event. The extracted measurements are then divided into 4 consecutive intervals of 90 minutes each. Additional columns include binary flags to identify whether a measurement is real, forward filled, or imputed, producing a final input tensor of 4 x 55, where 4 is the number of intervals and 55 the total number of features including the flags. Models such as XGBoost, GRU, and LSTM were evaluated using the processed dataset, and then further extended the GRU model by implementing a physiological branch constrained by the Alveolar Gas Equation and the Arterial Oxygen Content Equation. Synthetic episodes were also generated based on the training split using TimeDiff and evaluated for statistical fidelity and tail preservation.

After evaluation on the held-out test set from MIMIC-IV, the baseline and physics-informed GRU models have achieved AUROC scores of 0.6017 and 0.6021 respectively, as well as AUPRC scores of 0.136 and 0.135 respectively. Even though predictive capability remained modest, the physiological constraints implementation has significantly reduced the residuals associated with the 2 pulmonary gas equations from 33.267 to 0.208, and from 0.360 to 0.052. Other improved metrics include recall score from 0.178 to 0.373, and F1 score from 0.161 to 0.208. Additionally, despite slight improvements produced by the synthetically generated data, such as Kolgomorov-Smirnov and Wasserstein distance through capping at the extreme values, the synthetic augmentation did not improve predictive capability of the baseline model, and was therefore not further used to train the PINN model. Lastly, external validation was conducted using an external cohort produced from the eICU-CRD dataset with a total of 13,595 extubation episodes. The external validation scores resulted in an AUROC of 0.557, and an AUPRC of 0.083, therefore indicating limited performance preservation across separate datasets. Overall, physiologically constraining the model using the 2 pulmonary equations has improved physiological consistency of the output, but did not improve predictive capabilities.

Chapter 2

Declarations

2.1 Ethics and Data Governance

This project involves a secondary analysis of deindentified eletronic health records routinely collected in the settings of Intensive Care Unites (ICUs), and did not involve any direct intervention and/or interaction with the underlying patients. The explored datasets, such as MIMIC-IV and eICU-CRD were both distributed through PhysioNet under required credential access and established data use agreements [1], [2], where access was granted once the required training process was complete. Similarly, the AmsterdamUMCdb dataset was also provided under the same circumstances but with a separate application to the owners [3] as well as recommendation from Dr. Murali. Lastly, a curated and deidentified Royal Brompton Hospital dataset was provided by Dr. Murali.

Even though the health records have been deidentified, sensitive care was applied throughout the entirety of the project. All datasets were stored on a password protected computer, with the Royal Brompton Dataset being solely stored on Imperial College London's secure cloud storage called Box. Furthermore, only the necessary variables required by the project's goals were extracted, and presented aggregate characteristics throughout the report.

The synthetically generated extubation episodes were generated using the TimeDiff model, which has been trained solely on training split derived from the MIMIC-IV dataset. Synthetic data was not automatically assumed to be anonymized because generative models may overfit and reproduce exact representations of the real patient data, and statistical similarity does not establish data privacy protection by itself [4]. Furthermore, synthetically generated episodes were solely used for experimental training purposes, and were not published, shared, nor presented to unauthorized individuals to preserve patient privacy of the source training data.

Lastly, the models developed as part of this project were evaluated in retrospect, and not used in any clinical decision process. That is because prediction errors, as explored in more detail in future sections, could cause various clinical consequences, such as falsely predicting a successful extubation could lead to premature extubation, while a false prediction of an extubation failure could lead to prolonged ventilation or reintubation. Practical use of the developed models in this project would require further external validation, subgroup evaluation, as well as appropriate clinical oversight by a medical professional.

2.2 Use of Generative AI

Use of generative artificial intelligence (AI) tools included assistance from ChatGPT as a learning resource. It was used for proofreading for a limited number of report text to ensure the correct message is transmitted to the reader, while also being used to facilitate the understanding of various clinical terminology and clinical relationships, which are outside of my personal area of expertise. It was also used to facilitate a translation of tables containing results from coding experiments into LaTeX table formats. Lastly, it was used to assist with some printing blocks in order to extract information in a more organized and readable way, as well as in the proper conversion of

the extracted training data to an input tensor suitable for the TimeDiff model without disclosing any patient information or underlying values from any dataset, and have received the Supervisor's approval for this purpose in order to ensure task deadlines are met and no additional and unnecessary overhead is added.

ChaptGPT's outputs were thoroughly verified and treated as suggestions rather than primary sources. For example, a moment when it has proven to be unreliable was by inaccurately providing clinical information in regards to the equivalency of features such as peripheral oxygen saturation and arterial oxygen saturation, which verified and clarified by Dr. Murali.

2.3 Sustainability

The vast majority of model development, training, and evaluation was performed locally using an NVIDIA RTX 4060 GPU with additional GPU resources provided by Imperial College London's Data Science Institute, which have been used for a small number of days, and not extensively. The experiments used a 4 time step per episode structure, and no large GPU cluster or distributed training resources were required to accomplish the project's goals. Lastly, checkpoints were used to avoid future unnecessary retraining, and the carbon footprint of the project was not recorded because energy consuption was not closely monitored, but would have not been expected to result in a large amount.

2.4 Project Code Repository

The project code can be found at the following GitHub Repository: Link

Please note that the project was developed locally without consistent GitHub pushing due to personal preference style of work, and have received supervisor permission for the mentioned work style.

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Chapter 3

Introduction

3.1 Project Introduction and Motivation

Intensive Care Units (ICUs) provide coninuous treatment and monitoring for severely ill patients [5], with a significant proportion of these patients requiring invasive mechanical ventilation due to the inability to maintain adequate oxygenation independently [6]. Despite the life saving nature of Mechanical Ventilation (MV), prolonged exposure could lead to complications such as pneumonia, barotrauma, and aiway injury [7], while extubating a patient prematurely can result in extubation failure, leading to either reintubation, or being placed under non-invasive respiratory support. Therefore, risk management is required to avoid potential complications.

Mechanical ventilation weaning refers to a gradual decrease in ventilatory support, while patient extubation refers to the removal of the invasive respiratory apparatus, such as the endotracheal tube once a patient is considered fit to withstand spontaneous breathing trials. A medical professional may assess extubation readiness based on clinical information such as respiratory rate, oxygenation, neurological status, and ventilator settings to name a few. However, ventilation weaning is a complex process and no singular variable is capable of fully representing the patient's clinical condition [8]. This project's aim is to define the problem as an extubation failure prediction using features monitored and recorded within the final 6 hours prior to an extubation event, where extubation failure was defined as either reintubation within 48 hours after the extubation event, or prolonged non-invasive respiratory support for longer than 48 hours post extubation.

Electronic health records collected in ICU settings consist of several physiological measurements that are routinely collected and do not necessarily represent one specific illness, which makes machine learning an appropriate tool to identify the relationships that may aid the predictive task of ventilation weaning. Previous works have examined both predictive machine learning and time-series neural networks for patient extubation readiness [9], [10], [11], [12]. This project's aim is to analyze how much model architectures such as XGBoost, GRU, and LSTM can benefit from data augmentation and physiologically grounded constraints being incorporated into the predictive logic.

Due to a severe class imbalance, where extubation failures represented the minority class, a TimeDiff generative model was explored as a measure to address the issue though the synthetic generation of extubation failure episodes [13]. This data augmentation component investigated whether supplemental synthetically generated extubation episodes could improve the predictive performance of the model.

Lastly, models that are highly dependent on data may learn statistical relationships in the feature space without guaranteeing consistency with respiratory physiology. Therefore, the alveolar gas and arterial oxygen content equation were incorporated in the form of a physics-informed neural network to examine whether physiologically grounded constraints could improve predictive capability of the model [14], [15], [16]. Lastly, the predictive performance and physiological equation based residuals were evaluated separately, and a full PINN model comprising of all implementations was then externally evaluated on the eICU-CRD dataset.

3.2 Objectives

The primary aim of this project was to develop and evaluate a Physics-Informed time-series model for patient extubation readiness prediction in an ICU setting. In addition to the downstream prediction task, the project also explored whether synthetic data augmentation of extubation episodes could address the severe class imbalance, and whether incorporating pulmonary gas exchange equation based constraints into the model's logic could improve its predictive capabilities and physiological consistence for more explainable outputs. The final model's performance was assessed through an external validation procedure, which aimed to explore transportability to another ICU dataset.

The objectives consisted of:

1. The construction of a reproducible extubation episode cohort from MIMIC-IV and eICU-CRD with a defined binary label for extubation failure.
2. To transform the 6 hours decision window before an extubation event into 4 chronologically ordered time steps of 90 minutes each, consisting of dynamic and static physiological measurements, ventilator settings variables, as well as binary variables indicating the originating source of the corresponding value.
3. To establish baselines using only the direclty measured samples using models such as XGBoost, GRU, and LSTM.
4. To train and evaluate a generative TimeDiff model to determine whether synthetic data augmentation to address severe class imbalance could improve the predictive performance of the underlying models.
5. To incorporate physiological constraints based on the alveolar gas and arterial oxygen content equations, and evaluate both model predictive capability and physiological residual consistency with real measurements.
6. To compare the full PINN model against its corresponding baseline using patient bootstrap resampling, and externally evaluating the models using the cohort extracted from eICU-CRD.

Together, the objectives mentioned above examined 3 approaches of the extubation readiness prediction task, consisting of learning whether synthetically generated samples can facilitate relationships found in real data, address the class imbalance through various supplemental methods of synthetic data to real data, and physiologically constraining the learned representations by grounded pulmonary gas exchange equations.

Chapter 4

Literature Review

4.1 Mechanical Ventilation Weaning Prediction

The prediction of ventilator weaning readiness in the context of intensive care units (ICU) remains a critical and challenging problem to solve, especially when it comes to maintaining interpretability and generalization. The vast majority of recent Machine Learning studies on this topic address a similar problem but often vary in terms of prediction task, dataset, and occasionally in terms of methodology and prediction target.

Park et al.'s work addressed the prediction of ventilator weaning outcomes using only continuous monitoring parameters obtained from ventilators during Spontaneous Breathing Trials (SBTs) [9]. This is relevant to our project because it confirms that both waveform and numerical time-series ventilator parameters can be used to extract valid patterns in corelation with ventilator weaning by an engineered artificial neural network, which is a custom MobileNetV3 (CNN) architecture. However, this work focuses on patients who have already been deemed fit by a medical professional to undergo SBTs, which is a different problem formulation from our project. The authors mention crucial limitations such as having used a small, single-center, and self-collected dataset with an imbalanced sample distribution across the binary class architecture, which may further lead to overfitting, a high risk of false negative (FN) predictions, a limited number of applicable model architectures, and generalizability through external validation. Additionally, no physiological constraints have been implemented into the model's logic, which leaves open the possibility of learning more clinically accurate connections between ventilator parameters and ventilator weaning prediction, hence further motivating our work.

Zappala et al.'s work addressed the development and validation of a bedside Machine Learning clinical decision support tool to predict invasive mechanical ventilation (IMV) weaning readiness [17]. This is relevant to our project because it formulates the problem as a continuous, dynamically updating prediction, which is clinically accurate, as patient EHR data is time-series data. This work also addresses problems highlighted in Park et al.'s study, such as increasing the number of training samples to 8,565 extracted from the MIMIC IV dataset, and externally validating the models' performance on 2,626 samples extracted from the AmsterdamUMCdb dataset. Although the authors suggest that errors in missing data imputation in their hourly-updated data approach are mitigated using a forward-filling methodology [17], it may not prove as effective when it comes to rapidly changing ICU variables, such as respiratory rate, fraction of inspired oxygen, and ventilator settings, to name a few. Additionally, other critical limitations, such as a lack of performance comparisons with other off-the-shelf or custom neural network models, heavy emphasis on feature engineering, reaching a total of 2,992 features per IMV hour across the 3 models, and a lack of physiological constraints being implemented in the models' logic. Therefore, our project is further motivated by these limitations.

Huang et al.'s work addressed the development of a Machine Learning model for predicting successful patient extubation using only time-series ventilator-derived parameters [10]. This is relevant to our project because it supports that ventilator parameters such as expiratory tidal volume (Vte), respiratory rate (RR), peak airway pressure (Ppeak), mean airway pressure (Pmean), positive end-

expiratory pressure (PEEP), and fraction of inspired oxygen (FiO_2) contain relevant patterns in correlation with patient extubation outcomes given their proposed 3-minute sampling methodology, achieving results of $F1 \approx 0.95$ with an off-the-shelf Random Forest model [10]. While, the authors have identified the severe 78% positive to 12% negative imbalance in the sample distribution across the binary class architecture, and applied the synthetic minority over-sampling technique (SMOTE) successfully with minimal performance changes, other critical limitations remain at play. First, the dataset used is a small, single-cohort, and self-collected dataset [10]. Second, the only ventilator weaning mode taken into account was pressure support ventilation (PSV) [10]. Lastly, no physiological constraints have been implemented into the model's logic, which further motivates our work.

Xu et al.'s work addressed the development of a risk prediction model with comprehensive and individualized offline programs based on machine learning techniques aimed to help identify patients at risk of difficult weaning in time for early intervention to be facilitated [18]. This is relevant to our project because it supports the idea of ventilator weaning being a far more complex process with varying levels of difficulty and several edge cases that require early identification and patient-specific intervention. The problem is formulated to avoid the simple binary classification problem proposed in prior work. The authors support that patterns relevant to mechanical ventilation weaning difficulty can be extracted from 36 features across 4 categories: demographic, clinical factors, blood gas analysis parameters, ventilator parameters, and laboratory variables [18]. The proposed method achieves results of $F1 \approx 0.83$ with an off-the-shelf Random Forest Classifier model on a prediction target of difficult weaning when the patient is failing the first SBT, or needs up to 3 SBT episodes for ventilator weaning, or requires up to 7 days to be weaned successfully. However, there are several limitations at play. First, the dataset used is small and single-center, with 487 total samples and an imbalanced sample distribution across the binary class architecture, where 323 samples are classified as simple weaning and 164 as difficult weaning, which leads to increased bias. Furthermore, the dataset lacks diversity in the sample population in terms of age, gender, and socioeconomic backgrounds, which further increases bias risk and reduces generalizability [18]. Second, no custom neural architecture design is proposed nor compared during this study. Third, the authors do not emphasize a continuously updating prediction using ICU time-series data, but rather a static risk stratification prediction model, which differs from our work. Lastly, no physiological constraints have been implemented into the model's logic, which further motivates our work.

Qiu et al.'s work addressed the development and validation of an interpretable machine learning model that comprehensively evaluates the factors influencing ventilator weaning to provide clinical decision support for ventilator weaning in an ICU setting [19]. This is relevant to our project because the LASSO-based dimensionality reduction and statistical analysis conducted by the authors confirm the significance of 24 features from a multidimensional clinical space that includes demographics, comorbidities, vitals, treatment measures, blood gas parameters, blood test results, biochemical indicators, and coagulation parameters to contain important patterns for ventilator weaning prediction. Additionally, the authors apply the SHapley Additive exPlanations (SHAP) approach to rank contributing features for better interpretability, and to address significant limitations of other prior works discussed above. For example, they address the external validation issue identified in [9], [10], and [18], and the limited model performance comparison met in [9]. However, there are still significant limitations related to this work. First, even though a separate dataset is utilized for external validation purposes, it was collected within the same Hospital Affiliation System, which reduces the external validation analysis due to inherent systemic bias, potentially overlapping demographics of patients, and other institutional biases such as protocols and medications. Second, "gold standard" invasive monitoring methods for assessing a patient's volume status for critically ill patients, such as central venous pressure (CVP), pulmonary capilarry wedge pressure (PCWP), and pulse indicator continuous cardiac output (PiCCO) have been omitted from the study [19]. Third, the dataset used is small, with 1,371 total samples and an imbalanced sample distribution across the binary class architecture, where 85.12% of samples are classified as successful weaning and 14.88% as failed weaning, leading to increased bias. Lastly, no physiological constraints have been implemented into the models' logic.

Sheikhalishahi et al.'s work explored the application of machine learning to predict successful weaning from MV, focusing on the positive end-expiratory pressure (PEEP), a key component of MV, as a potential predictor for successful weaning [11]. While the problem is not formulated

for ventilator weaning specifically, it analyzes PEEP reduction, which is a crucial component in the process of mechanical ventilation weaning. This is relevant to the project because ventilator weaning is a gradual process that involves progressive changes in the ventilator settings to reduce dependency, which later lead to respiratory stability tests such as SBT, and PSV to name a few, and ultimately to extubation, which also requires close post-extubation monitoring. Therefore, ventilation weaning is not treated as a simple binary classification problem and, as mentioned by the authors, a PEEP-level of 8 cmH2O or less is an indicator for "minimal dependency" and one of the required criteria for weaning from mechanical ventilation [11]. This study also addresses limitations highlighted in previous works. For example, they address the external validation issue identified in [9], [10], [18], and [19], and the small dataset limitation identified in [9], [10], [18], and [19] by extracting a total of 12,153 samples, and utilizing institutionally and geographically distinct datasets for training and testing of the models, such as the MIMIC-IV dataset, and for external validation, such as the eICU dataset. While the authors do not specify limitations other than disease-specific applications of the models [11], there are several that can be identified. First, while data missingness is handled by forward-imputing, temporal trajectories are not specifically modeled as part of the approach. Second, given the continuously updating nature of MV patients in ICU settings, this study does not provide an updated prediction outcome based on clinical time-series data. Third, in the case of both datasets, with 10,700 total samples from MIMIC-IV and 20,162 total samples from eICU, the sample distribution is imbalanced across the binary class architecture, where 70% and 35% of samples are classified as positive, respectively, leading to increased bias. Lastly, no physiological constraints have been implemented into the models' logic.

Table 4.1: Comparison of Machine Learning Prior Papers for Ventilator Weaning

Paper	**Prediction Task**	**External Validation**	**Decision Window**
Park et al. [9]	Weaning Success vs. Weaning Failure	NO	48 hours
Zappala et al. [17]	Weaning Success	YES	24 hours 48 hours 72 hours
Huang et al. [10]	Extubation Failure	NO	3 Minutes
Xu et al. [18]	Difficult Weaning	NO	Before SBT
Qiu et al. [19]	Weaning Success	YES	NOT Specific. When recent, last 24-hour, vitals, lab test results, blood gas parameters, demographics, and comorbidities available
Sheikhalishahi et al. [11]	Peep Reduction Success vs. Failure	Partial	Event-based

4.2 Physics-Informed Machine Learning

Zhao et al.'s work does not specifically address prior work limitations, or introducing a novel architecture, but rather comprising of a review comparing more than fourty physics-informed neural network past papers focused on biomedical and physiological signal processing applications [16]. This is important for the project because it provides a comprehensive breakdown of all the relevant components of physics-informed neural networks such as the physical models, loss function, data

preprocessing, as well as the activation function, and the most widely used implementation choices for each of the specified components. For example, Zhao et al. specify that hyperbolic tangent, sigmoid, and ReLU comprise of approximately 77% of the activation function choices in PINNs [16]. Additionally, Zhao et al. provide information about the expected training time, which can reach up to $\approx$ 110 hours, most common data types, datasets, optimizers, as well as evaluation metrics such as the mean squared error, the averaged mean squared error, the root mean squared error, and the relative error, to name a few. Therefore, Zhao et al.'s work further motivate the incorporation of physiological constraints in model logic by providing a comprehensive literature review of key PINN prior works focused on biomedical and physiological applications.

Sarabian et al.'s work addressed the development of a physics-informed deep learning framework aimed at augmenting sparse clinical records in order to generate clinically accurate time-series brain hemodynamic parameters [20]. Although the problem is not specifically formulated for ventilator weaning, it highlights how physiological constraints based on reduced-order single dimensional pulsatile blood flow models improve the prediction performance of brain hemodynamics, which further motivates applications for cerebrovascular disease. Additionally, the authors developed an Area Surrogate Physics-Informed Neural Network (ASPINN) meant to adress the lack of critical variables in time-series format, specifically the blood flow velocity, blood pressure, and vessel cross-sectional area [20]. This is relevant to the project because, depending on what hospital system the dataset has been collected under, underlying variables necessary to integrate physiological constraints into the model logic may not be directly available, or lacking temporal frequency and alignment. However, as mentioned by Sarabian et al., several limitations based on simplified physiological modeling and assumptions are identified, such as neglecting complex pressure distribution, three-dimensional effects of vessel tortuosity and curvature as a result of chossing an axisymmetrical model, among others [20]. This is important because similar assumptions and model simplification approaches may be necessary when implementing pulmonary gas exchange physiological constraints.

Sel et al.'s work addressed the establishment of a Physics-Informed Neural Network aimed to extracting cardiovascular parameters from physiological time-series data, while limiting the use of ground truth data [21]. While this work's problem is not focused specifically on ventilator weaning, it highlights how Taylor approximation can be utilized to implement physiological loss terms into the model logic to aid in estimation of cuffless blood pressure based on wearable bioimpedance data. This is important to the project because even though no well-established physiological equation is incorporated into the model logic, the residual loss is based on the Taylor approximation of the input features and the blood pressure data, therefore supporting the idea of physiological constraints while lacking ground truth data in time-series models. However, aside from the limitations identified by the authors, such as a small cohort size dataset and the model calibration lacking real-world feasibility [21], another significant limitation is the lack of implementing a physiological equation into the loss function, which would require significant adaptations and structural changes of this approach due to the underlying variables required to compute the equation.

Prior to a full integration of the pulmonary gas exchange model, while awaiting permission to access it, an initial feasibility analysis is conducted to assess whether the chosen datasets can support this implementation, and to what degree variables will need compensation from the generative component. This is relevant because it focuses on identifying the variables required to implement each of the four clinically well-established gas exchange equations, and mapping these to features available in the chosen datasets. Additionally, the four highlighted equations, which are described in more detail later in this section, will constitute the preliminary implementation experiments aimed at integrating clinically accurate physiological constraints into the model logic.

This initial stage revolves around four clinically established gas exchange equations: The Alveolar Gas Equation, The Arterial Oxygen Content Equation, The Shunt Equation, and the Fick Principle. These equations sit at the base of today's clinical practice in the ICU and critical care examinations, providing a clinically relevant starting point for developing the Physics-Informed component of this project. This is important because not only it avoids progress delay associated with the permission to access the pulmonary gas exchange model, but also because it helps identify physiological constraints that can be integrated based on what variables are available directly, can

be derived from other directly available variables, or require representation through latent variable estimation.

The Alveolar Gas Equation (4.1) is used to compute the partial pressure of oxygen in the alveoli. This is important because, according to Sharma et al., it is impossible to collect gases directly from the alveoli, but respiratory quantities such as fraction of inspired oxygen and pressure of carbon dioxide in arterial blood are commonly recorded in ICU datasets. Furthermore, while not as commonly measured, variables such as atmospheric pressure, partial pressure of water, and the respiratory quotient are based on standard clinical assumptions of 760 mm Hg at sea level, ≈ (45) mm Hg, and ≈ (0.82) for the human diet average value, respectively [14]. This is relevant to the project because the equation provides a clinically accurate physiological constraint based on the relationship between its parameters, incorporating logic that penalizes outputs, which violate the expected behavior.

$$P_{AO_2} = [(P_{atm} - P_{H_2O})FiO_2] - \frac{PaCO_2}{RQ} \tag{4.1}$$

Where:

- P_AO_2: Avleolar Oxygen Partial Pressure
- P_{atm}: Atmospheric Pressure
- P_{H_2O}: Partial Pressure of Water
- FiO_2: Fraction of Inspired Oxygen
- P_aCO_2: Partial Pressure of Carbon Dioxide in Arterial Blood
- RQ: Respiratory Quotient. The amount of Carbon Dioxide (CO_2) Produced Divided by The Amount of Oxygen Consumed

The Arterial Oxygen Content Equation (4.2) is used to compute the total volume of oxygen carried in the arterial blood. This is important because Rhodes et al. highlight how the efficacy of oxygen transportation is critically related to parameters such as the hemoglobin concentration, partial oxygen pressure, and the hemoglobin saturation [15]. This is relevant to the project because by incorporating the underlying relationship of this equation into the model logic, it will support clinically accurate representations with clear physiological constraints based on the difference between oxygen bound to hemoglobin and oxygen dissolved in plasma [15]. Additionally, relevant parameters such as hemoglobin concentration, partial pressure of oxygen, concentration of oxygen, and hemoglobin saturation are commonly recorded measurements in the ICU, which further supports the physics-informed implementation approach of this equation.

$$CaO = 1.34 * [Hgb] * (SaO_2/100) + 0.003 * PaO_2 \tag{4.2}$$

Where:

- CaO: Concentration of Oxygen in the Blood
- Hgb: Hemoglobin Concentration
- SaO: Hemoglobin Saturation
- PaO_2: Partial Pressure of Oxygen

The Shunt Equation (4.3) is used to compute the fraction of cardiac output that bypasses the alveoli, therefore passing through the lungs without being involved in gas exchange [22]. Toffaletti et al.'s work highlights the idea that the shunt cannot be measured directly, but it is a measurement that is rather inferred from differences between end-capillary oxygen content, arterial oxygen content, and mixed venous blood [22]. This is relevant to the project because by incorporating the underlying relationship of this equation into the model logic, it will support clinically

accurate representations with clear physiological constraints based on oxygenation failure. However, as mentioned by the authors, despite the critical nature of underlying parameters, datasets lack recordings of these quantifications, specifically end-capillary oxygent content, arterial oxygen content, and mixed venous oxygen content [22]. Therefore, this limitation is affecting a physics-informed approach implementation.

$$\dot{Q_S}/\dot{Q_T} = (C_{c'O_2} - C_{aO_2})/(C_{c'O_2} - C_{\bar{v}O_2}) \tag{4.3}$$

Where [22]:

- $\dot{Q_S}$: Shunted Blood Flow
- $\dot{Q_T}$: Total Cardiac Output
- $C_{c'O_2}$: End-Capillary Oxygen Content
- C_{aO_2}: Arterial Oxygen Content
- $C_{\bar{v}O_2}$: Mixed Venous Oxygen Content

The Fick Principle (4.4), also known as the Inverse Fick Equation, is used to compute the cardiac output [23]. This is relevant to the project because by incorporating the underlying relationship of this equation into the model logic, it will support clinically accurate representations with clear physiological constraints between oxygen consumption, cardiac output, arterial oxygen concentration, and venous oxygen concentration. However, similarly to the Shunt Equation (4.3), this equation's variables may not be among the commonly recorded data items, and therefore suffer an implementation limitation of the physics-informed approach.

$$CO = VO_2/(C_aO_2 - C_vO_2) \tag{4.4}$$

Where:

- CO: Cardiac Output
- VO_2: Oxygen Consumption (ml/min)
- C_aO_2: Arterial Oxygen Concentration
- C_vO_2: Venous Oxygen Concentration

4.3 Synthetic EHR Data Generation

Zhong et al.'s work addressed the development of a diffusion-based model designed to predict the next visit based on the current one, while also incorporating interval estimation, a novel time-aware visit embedding module, and a pioneering predictive denoising diffusion probabilistic model (PDDPM) [24]. This is relevant to our work because ventilator weaning relies heavily on time-series variable progression rather than on static, hourly updates. The authors specifically address the limitations of prior works, such as developing a custom architecture with a novel time-aware visit embedding module rather than relying on the off-the-shelf generative techniques like generative adversarial networks, variational autoencoders, and language models, which provide limited time-series data generation due to the inherent approach of input replication [24]. Additionally, other addressed limitations include failure to simulataneously generate time intervals between visits, limited capability in learning visit representations, and the lack of a robust genereation model to balance data diversity and quality [24]. However, even though time-awareness is introduced in the generation process, it is focused on predicting the next event and its associated time interval, rather than generating frequently updating ICU time-series data. Furthermore, other underlying limitations, such as not specifically assessing whether the EHRPD model overfits the original, sensitive patient data, and lacking downstream task evaluation to evaluate model performance on synthetically generated data. Lastly, no physiological constraints are implemented in the model's

logic to ensure clinically accurate variable ranges rather than being bounded by existing data.

He et al.'s work addressed the challenge of generating both static and temporal heterogeneous Electronic Health Records (EHRs) jointly [25]. The authors also identify and address critical limitations, such as the missing correspondence in the data, by formulating an optimal transport problem in the embedding space to enable the construction of meaningful latent embedding pairs, and maintaining high-quality synthetic data for instances where modality is missing [25]. This is relevant to our work because it supports synthetic record generation in ICU settings, which naturally contain heterogeneous data, such as ventilator settings, demographics, vitals, laboratory test results, blood gas parameters, and comorbidities to name a few. Additionally, the proposed FLEXGEN-EHR model offers a better alternative to simple forward-imputation of missing records, which allows it to be trained on datasets with missing entries for modalities such as laboratory test results, blood gas parameters, etc. However, several critical limitations are at play. First, even though datasets with missing records is addressed, no physiological constraints are implemented in the model's logic to ensure clinically accurate variable ranges rather than being bounded by existing data. Second, although the authors have evaluated the model against membership inference attacks, it is still susceptible to privacy risks such as re-identification or membership inference attacks by inadvertently encoding sensitive information in the generated data.

Yan et al.'s work does not specifically address prior work limitations, but rather provides a comprehensive analytical tutorial focused on synthetically generating structured EHR data, comparing existing Generative Adversarial Networks (GANs) studies, data preprocessing, and evaluation. This is relevant to our work because it covers the pipeline for downstream clinical outcome and support tasks, including critical preprocessing steps such as removing outliers, normalizing continuous variables, handling missing values, and handling concepts with low prevalence, which correspond to clinical variables that represent rare facts or events within the patient cohort [26]. Additionally, the authors highlight the importance of synthetic data quality evaluation to ensure data utility, patient privacy, and fairness through comparison between synthetic and real data using metrics such as dimension-wise distribution, column-wise correlation, latent cluster analysis, medical concept abundance, clinical knowledge violation, prediction performance, and feature importance ranking [26]. However, there are several limitations associated with this study. First, it focuses on stimulating static structured EHR data and neglects the timestamping of medical events, while EHR data inherently consists of temporal information that is critical for numerous applications, including ventilator weaning prediction [26]. Second, the study lacks ability to capture complexity inherent in real EHR data [26]. Lastly, no physiological constraints are implemented in the model's logic to ensure clinically accurate variable ranges and values, which leads to missing on edge cases and generating clinically impossible data.

Table 4.2: Comparison of Machine Learning Prior Papers for Synthetic EHR Data Generation

Paper	Dataset	Type of EHR Data	Performance Preservation
Zhong et al. [24]	MIMIC-IV Breast Cancer Trial	Diagnosis, Drug, Lab Items, Procedures, Adverse Events, Medication, Lab Category, Treatment	Best in 7/8 modalities across 2 datasets against the baselines and according to relevant metrics: Longitudinal Imputation Perplexity (LPL) and Cross-Modality Imputation Perplexity (MPL)
He et al. [25]	MIMIC-IV eICU	Heterogeneous Medical Records (static and longitudinal)	Best against all baselines in prediction of mortality and acute respiratory failure (ARF) tasks.
Yan et al. [26]	MIMIC-IV	Demographics, Timestamped Medical Event, Medication Prescriptions, Measurements, Medical Procedures, and Clinical Outcomes	Decent, but limited in clinical consistency. Presented poor results in 10x more likely to wrongfully assign male diagnosis to female patients.
Tian et al. [13]	MIMIC-III MIMIC-IV eICU Stocks Dataset EnergyDataset	EHR time-series (continuous and discrete) and non-EHR time series.	Best against all baselines in prediction of mortality

4.4 Data

This project will examine and analyze four cohort-distinct intensive care datesets: the Medical Information Mart for Intensive Care IV (MIMIC-IV) v3.1, the electronic Intensive Care Unit Collaborative Research Database (eICU-CRD or eICU) v2.0, Amsterdam University Medical Centers Database (AmsterdamUMCdb) v1.0.2, and a respiratory-focused dataset obtained from the Royal Brompton Hospital (RBH). These datasets, and not limited to, were selected based on key factors, such as originating from ICUs across distinct institutions, and collected in different healthcare systems and data collection settings. The primary goal is to utilize the RBH Dataset for external validation given its United Kingdom origins and well-defined structure.

MIMIC-IV is a large, deidentified dataset of patients admitted to the emergency department or an ICU at the Beth Israel Deaconess Medical Center in Boston, MA between 2008 and 2022 [1]. It contains data for over 65,000 patients admitted to an ICU, and for over 200,000 patients admitted to the emergency department [1]. The schema is split into two modules: hospital and ICU. The hospital module contains information such as patient demographics, hospitalizations, intra-hospital transfers, laboratory measurements, microbiology cultures, provider orders, medication administration, prescriptions, and online medical record data, while the ICU module contains intravenous and fluid inputs, ingredients for the aforementioned inputs, patient outputs, procedures, information documented as a timestamp, and other charted information [1]. MIMIC-IV is relevant to this project because it contains time-series ICU data that can be used to extract ventilator variables, laboratory results, clinical outcomes, and blood gas parameters, to name a few.

The eICU, or eICU-CRD, is a multi-center database containing deidentified health data associated with over 200,000 patients admitted to one of 335 ICUs at 208 hospitals across the United States of America between 2014 and 2015 [2]. It contains relevant information, such as vital sign measurements, care plan documentation, APACHE severity of illness measures, and diagnosis information and treatment details [2]. eICU is relevant to this project because it provides data collected in a multi-center structure, which helps assess model generalizability and evaluate perfor-

mance beyond single-cohort hospital systems. However, the authors highlight that different care units may utilize distinct data collection interfaces, and a lack of common interface will result in data missingness for given patients [2].

The AmsterdamUMCdb is a European intensive care database endorsed by the European Society of Intensive Care Medicine (ESICM) and its Data Science Section. It contains deidentified health data related to approximately 23,106 ICU and high dependency unit admissions of adult patients between 2003 and 2016 [3]. AmsterdamUMCdb is relevant to this project not only because it offers another hospital system data collection to train and evaluate the model on, but also because it provides a European ICU data source to compare with USA-derived datasets such as MIMIC-IV and eICU-CRD. Murali et al.'s work highlights that invasive mechanical ventilation practice and weaning are are significantly influenced by international differences in the clinical workflow, such as clinical organization, staffing, protocols, and the responsibility of nurses and physicians in the decision-making process [27].

The RBH dataset has been curated by Dr Mayur Murali, and for which access permission has been granted. This is relevant to the project because the Royal Brompton Hospital is a specialist heart and lung hospital, and the dataset is deidentified and contains relevant information such as demographics, vital signs, ventilation variables, laboratory test results, blood gas parameters, and vasopressor information, to name a few. Additionally, the RBH dataset provides another international perspective to a hospital system regarding clinical protocols, medication, and data collection systems, originating from the United Kingdom. Lastly, this dataset is intended to be used for the external validation of all three project components. The dataset is securely stored on Box, which is a secure cloud-based storage and collaboration service for controlled file sharing among approved collaborators. The RBH dataset will not be transferred out from Box during the full length of this project, unless instructed otherwise by the supervisor.

Access to the publicly available datasets, such as MIMIC-IV, eICU-CRD, and AmsterdamUMCdb was granted after successfully completing the CITI Data or Specimens Only Research certification course and signing the appropriate data use agreements. Even though the datasets are deidentified, one must handle the data responsibly due to the clinically detailed nature of the information provided.

At this stage of the project, the datasets are analyzed to explore and assess suitability for the mechanical ventilation weaning prediction task. It is trivial that the datasets contain mechanical ventilation variables, ventilator settings, blood gas parameters, oxygenation variables, ICU stay information, and ventilated patient outcomes, to name a few. Furthermore, given the lack of access to the pulmonary gas exchange model at this point in time, the early experiments will be conducted using four established physiological equations that manage gas exchange: the alveolar gas equation, the shunt equation, the oxygen content of arterial blood equation, and the Fick principle, which were recommended by Dr Murali. These provide an initial clinically accurate framework to identify dataset suitability for the pulmonary gas exchange model implementation later on. The equations are discussed in more detail in the Physics-Informed Machine Learning section, but a suitable dataset will include variables such as fraction of inspired oxygen (FiO_2), partial pressure of arterial oxygen (PaO_2), partial pressure of arterial carbon dioxide ($PaCO_2$), and oxygen saturation, among others.

The choice of candidate data sources is motivated by the identified key limitations in past mechanical ventilation weaning studies of relying on single-cohort datasets, and prediction targets that do not resemble clinically accurate ICU decision-making. By analyzing and comparing multiple datasets that are different in terms of hospital system, ventilation weaning protocol, and country of origin will help identify what features are consistently available across various ICU settings, and what data preprocessing and pipeline methods are required to ensure model generalizability while maintaining a clinically meaningful process. Lastly, while datasets such as MIMIC-IV, eICU-CRD, and AmsterdamUMCdb provide large-scale sampling and reproducibility, the RBH dataset offers a clinically specialized perspective of cardiovascular and respiratory care.

An initial, clinically-grounded orientative list of variables compiled from the MIMIC-IV and

RBH Datasets has been provided by Dr Murali with the intent of assessing dataset suitability for the implementation of physiological constraints based on the Alveolar Gas Equation (4.1) and the Arterial Oxygen Content Equation (4.2) along with their underlying variables. Even though the list is tailored to the available tables, naming conventions, and units of measurement specific to MIMIC-IV, this list is comprised of commonly collected variables in ICU settings. However, given the distinct hospital systems where the proposed datasets originate from, significant limitations such as discrepancies in the units of measurement, table and variable naming conventions, temporal data recording convention, and potentially language.

Table 4.3: List of Variables derived from the MIMIC-IV Dataset, which are related to the Alveolar Gas Equation, Arterial Oxygen Content Equation, and the underlying variables.

Table	**Variable(s) (Unit)**
ICU stay	stay id, patient id, in time, out time, length of stay, icu mortality
Demographics	admission weight (kg), height (cm), estimated height, age (years), sex, comorbidities (as established)
Admission	APACHE II score, SOFA score, renal (creatinine), cardiovascular (MAP/ vasopressors), haematological, hepatic, neuro (GCS), admission diagnosis, admission source, elective or emergency
Ventilation	offset, hour timestamp, airway, FiO_2, ventilation mode, PEEP, tidal volume, positive inspiratory pressure, pressure support, peak airway pressure, plateau pressure, driving pressure, respiratory rate, minute volume, inspiratory-expiratory ratio, end-tidal carbon dioxide measurement, set respiratory rate, actual respiratory rate, secretion burden, color of sputum
Vital signs	heart rate, systolic blood pressure (mmHg), diastolic blood pressure (mmHg), mean arterial pressure (mmHg), temperature (C), oxygen saturations (%), respiratory rate, GCS, RASS score
Blood gas	pH, pO_2 (kPa), pCO_2 (kPa), BE, HCO3, lactate, sodium, chloride, potassium, calcium, haemoglobin, haematocrit, methaemoglobin, carboxyhaemoglobin, glucose, type
Laboratory	creatinine, urea, sodium, potassium, bilirubin, ALT, AST, ALP, platelets, INR, APTT, PT, creatine kinase, WCC, LDH, albumin, haemoglobin, haematocrit, MCV, magnesium, calcium, phosphate, CRP, procalcitonin, eGFR
Organ support	ventilation, NIV, IABP, ECMO, RRT, vasopressors, inotropes
Input / Output Events	all events based on the inclusion and exclusion criteria on hourly basis and total over 24 hours

Chapter 5

Methodology

5.1 System Design

This project followed an end to end workflow consisting of dataset suitability assessment, cohort construction, feature preprocessing, synthtetic data generation, predictive modelling, physics-informed model development, and external validation. The candidate datasets comprising of intensive care unit records were initially assessed based on whether ventilation episodes, patient extubation outcomes, and and the required feature space could be reconstructed reliably. MIMIC-IV was selected as the primary dataset for model development and internal evaluation, while eICU-CRD was selected for external validation.

After valid cohorts were constructed, all valid extubation episodes were transformed into 4 time step representations consisting of dynamic physiological measurements, other static features, as well binary variables indicating the source of the value. The MIMIC-IV cohort was split on a patient basis into training, validation, and held-out test splits. Models such as XGBoost, GRU, and LSTM were evaluated as predictive baselines, and then synthetically generated extubation episodes generated by a TimeDiff model were investigated in data augmentation experiments due to a severe class imbalance commonly met in the 2 extracted cohorts. Then, the selected GRU architecture was further expanded to including auxiliary physiological prediction states, and constraints based on the alveolar gas and arterial oxygen content equations.

Lastly, all preprocessing parameters, synthetic data generation procedure, and any model training were limited to the MIMIC-IV derived training set only to avoid evaluation sample leakage. The MIMIC-IV validation split was then utilized for model selection, hyperparameter fine tuning, and decision threshold derivation, while the held-out test split was solely reserved for internal performance evaluation. eICU-CRD was used as an external validation dataset to assess transportability of the selected physics informed GRU model with 2 physiological constraints, while using unchanged weights and decision threshold. The individual stages of this workflow are described in more detail in Sections (5.2 - 5.8).

5.2 MIMIC-IV and Cohort Construction

Based on the candidate datasets introduced in Section (4.4), MIMIC-IV version 3.1 was selected as the primary dataset to conduct the model development, internal validation, and model performance evaluation [1]. This section focuses on the construction of the patient cohort according to the inclusion and exclusion criteria being applied to the MIMIC-IV dataset.

The inclusion and exclusion criteria used for the project were initially adapted from a past work of Akram Yoosoofsah and Dr Murali [28], and then updated in consultation with Dr Murali for the adaptation to this project.

Inclusion:

- 1. The patient must be admitted to an Intensive Care Unit (ICU).

- 3. The patient must have undergone mechanical ventilation.
- 4. The patient must have been extubated.

Exclusion:

- 1. The patient's age is < 18 or > 89
- 2. Duration of ventilation < 24 hours or > 30 days.
- 3. Repeated hospital admissions of patients other past the first admission.
- 4. Patients under palliative care and/or end-of-life.

Additionally, in order to maintain a clinically accurate approach, each extubation episode's eligible time-series ICU data is extracted based on a sliding window of **6 hours** preceeding the extubation event, as advised by Dr Murali. The decision window is further divided into **4 consecutive time steps** of **90 minutes** each.

Lastly, this project will treat the ventilator weaning prediction task on a binary class basis. That is, **Extubation Success** and **Extubation Failure**, where

Extubation Failure:

- Patient is reintubated within 48 hours of extubation.
- Patient remains on non-invasive ventilation for longer than 48 hours post-extubation.

Otherwise, the label is **Extubation Success**.

The primary label of analysis focuses on extubation events and episodes. This is distinctive from trating each hospital admission episode, ICU stay, or dataset sample entry independently, because a unique patient may experience more than one episode of invasive ventilation and more than one extubation attempt events. Therefore, each extubation label was associated with a specific extubation event, and the corresponding episode of invasive ventilation that is preceding. Given that several episodes qualify as eligible from a patient's first hospital admission, a design choice was to retain all episodes that belong to the very same patient in the same partition. The complete process used to construct the valid cohort, including the extubation label can be found in Figure (5.1).

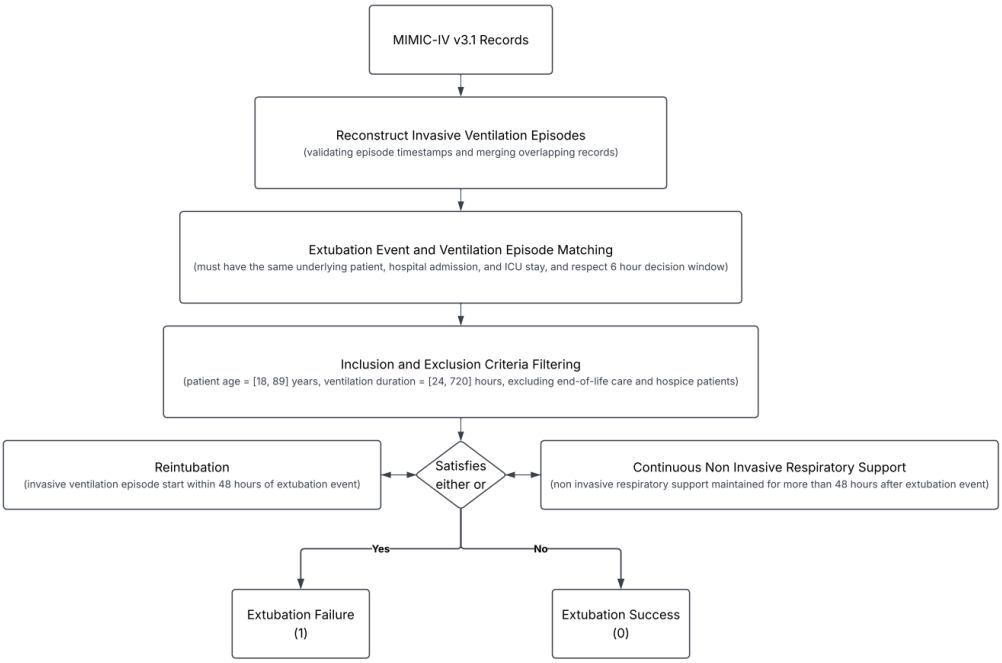


Figure 5.1: Diagram of the cohort-construction pipeline. It displays the pipeline steps, filtering, and labeling process of the MIMIC-IV v3.1 dataset.

The pipeline for constructing the cohort initially selects all the recorded ventilation start and end times to later reconstruct continuous episodes of invasive mechanical ventilation. All samples where the end time preceeds the start time have been removed, while samples where procedures overlap have been combined to avoid fragmental representation of a single episode as multiple separate episodes. Then, the selected ventilation episodes were required to also satisfy the criterion of lasting at least 24 hours.

Each selected extubation event was then linked in a backwards manner with the most recent invasive ventilation episode that belongs to the same underlying patient, hospital admission, and ICU stay. The value set at extubation time was required to match the decision window of 6 hours, meaning that the extubation event must occur no longer than the allowed 6 hours after the end time of the corresponding invasive ventilation episode. This method was implemented as a tolerance measure to account for small errors between the recorded end times of ventilation episodes and recorded times of the extubation events. Additionally, this method further reduces the risk of matching an extubation event to an unrelated ventilation episode that occured earlier in the timeline. Furthermore, any duplicated records that correspond to the same unique patient, hospital admission, ICU stay, and extubation time have been removed.

Eligible patient population has been selected based on the inclusion and exclusion criteria presented in Section (3.2) with the initial requirement of including only adults with ages between 18 and 89. Then, to reduce label influence caused by repeated hospital admissions, only the first admission of each patient was retained in the cohort, even though multiple extubation events could have been extracted from the excluded readmissions. Hospital admission samples where either hospice discharge or palliative care was explicitly stated have been exluded as per the inclusion and exlusion criteria. Additionally, patients recorded with death during a ventilation episode were not included either because, as advised my Dr. Murali, such an event may be caused by several underlying reasons that are unrelated to extubation readiness.

The selected invasive ventilation episodes that lasted for longer than 30 days (or 720 hours) have also been removed prior to the feature extraction step. This filter was applied as a design choice aimed to improve data quality rather than a clinically informed criterion. It prevented a single digit amount of samples with extreme ventilation duration values from skewing the distribution of

ventilation duration in the preprocessing step. Furthermore, later ventilation episodes lasting less than 24 hours still qualify to indicate reintubation events because the lower bound constraint only applies to the ventilation episode preceding the extubation event.

The extubation event label was then assigned on a binary basis for each selected ventilation episode. Extubation failure was assigned when either invasive ventilation reintubation occured within 48 hours after the extubation event, or non invasive respiratory support continues for longer than 48 hours past the extubation event. The non invasive respiratory support qualifiers included bilevel positive airway pressure (BiPAP) mask, continuous positive airway pressure (CPAP) mask, high-flow nasal cannula, the Hamilton DuoPAP, non-invasive ventilation (NIV), and non-invasive ventilation in spontaneous/timed (NIV-ST) modes. A label is successful if neither condition of the ones mentioned is satisfied, and unsuccessful if otherwise. Each samples should have then been provided a binary label based on the criteria discussed.

Lastly, any further episode information specific to after the extubation event occurred was used to ultimately determine the final outcome. In order to prevent clinical information leakage into the model input, any features selected for prediction were limited to recordings prior to the extubation event. The final cohort size, class distribution, and number of samples that have been impacted by the applied filters can be found in Section (6.1).

5.3 Dataset Suitability Assessment

Dataset suitability is assessed before the model development stage in order to determine which dataset has the ability to support the intended modeling experiments as well as potentially supporting external validation. Datasets such as MIMIC-IV, eICU-CRD, RBH, and AmsterdamUMCdb were initially considered against the same project requirements. the dataset assessment examines whether ventilation episodes and corresponding extubation events can be reconstructed, whether the defined inclusion and exclusion criteria can be applied, whether extubation outcomes for each episode can be derived consistently, and whether the required features are available withing the established pre-extubation decision window of 6 hours. A preliminary inspection of each dataset's schema, and clinical terminology was conducted before the quantitative feature space assessment.

The extubation episode and outcome suitability was assessed by analyzing the patient, hospital admission, and ICU stay IDs in addition to timestamps corresponding to ventilation, extubation, reintubation, and ventilation respiratory support provided after the extubation event. This analysis determined whether ventilation episodes could be matched to their corresponding extubation events, as well as whether the extubation failure definition defined in Section (5.2) could be properly constructed without ambiguity. The primary focus was to examine whether invasive and non-invasive ventilation could be distinguished from one another, and whether ventilation events that occur within 48 hours after an extubation event can be identified.

Feature suitability was then assessed by mapping each of the required dynamic and static physiological variables to its corresponding source table (or .csv file), as well as clinical label, and unit of measure. This analysis consisted of variables required by the alveolar gas and arterial oxygen content equations, because a physics-informed component could therefore not be implemented if the underlying features did not exist. While some variables are equivalent but stored under different names and associated with different units of measure, a set of rules for extraction and unit normalization had to be implemented on a dataset basis, which may also be required given that collection, hospital, and measuring systems can differ between countries, hospitals, and even ICUs.

On the feature level, each dynamic feaure's ratio of values originating from sources such as direct measurements, forward filling and interpolation, or MICE imputation were identified prior to conducting the preprocessing procedure described in Section (5.4). Furthermore, feature distributions were compared based on median, interquartile ranges, upper and lower bounds and percentiles in order to identify shifts and clinically impossible values that may have appeared during hamonization. Lastly, each dataset was examined on whether it could have been converted into 4 time step of 90 minutes each, 55 feature space representation, and one binary outcome variable to satisfy the predictive models' downstream task requirements. Further findings and results on a

dataset basis, as well as rationale for selecting the model development and external datasets are discussed in more detail in Section (6.2).

5.4 Feature Extraction and Data Preprocessing

The extracted extubation events and ventilation episodes constructed in Section (5.2) were preprocessed into fixed-lentgth time-series samples to act as input for the model development stage. The preprocessing pipeline consisted of 7 steps, which include the feature extraction, unit conversions and normalization, temporal aggregation of 6 hour episodes into 4 bins of 90 minutes each, filling missing values within the same episode, imputation using MICE, standardization and masking values as observed vs. mice imputed, and lastly, exporting the 4 x 55 input-ready data. A visualization of the preprocessing pipeline can be seen in Figure (5.2). Additionally, the preprocessing stage has only been using values prior to the corresponding extubation event occurred.

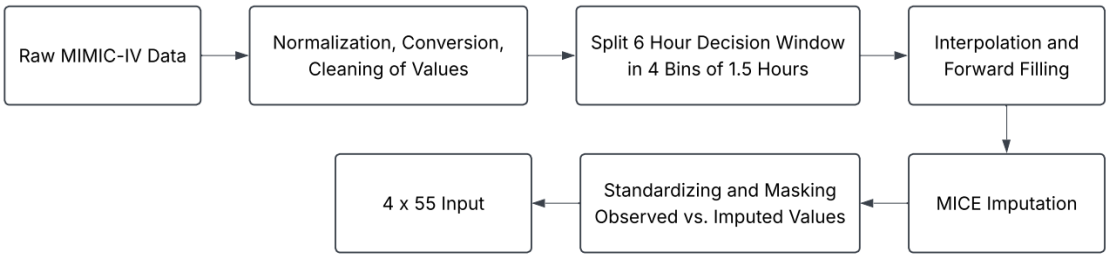


Figure 5.2: Diagram of the preprocessing pipeline. It displays the pipeline steps to achieve the model input data from the raw MIMIC-IV v3.1 cohort.

5.4.1 Feature Selection and Extraction

The feature selection stage focused on combining the variables required to satisfy the Physiological Equations (4.1 and 4.2), clinically relevant recommendations provided by Dr. Murali, practical availability within the selected dataset (MIMIC-IV), as well as evidence from past work on extubation prediction reviewed in Section (4.1). The ultimate feature set was not selected based on singular study, but rather explored relevant past works, such as [29], [10], [9], [30], and [12] to compile a final set based on identifying a repetitive set of features relevant to extubation prediction that includes physiological gas-exchange features, respiratory features, cardiovascular states, fluid balance, and ventilation.

The input-ready data contained 17 dynamic features selected to represent the physiological system related to ventilator weaning. Starting with the physiological features relevant to the Alveolar Gas (4.1) and the Arterial Oxygen Content (4.2) equations, these included the fraction of inspired oxygen (FiO_2), arterial partial pressure of carbon dioxide ($PaCO_2$), arterial partial pressure of oxygen (PaO_2), peripheral oxygen saturation (SpO_2), hemoglobin, and arterial pH. The respiratory mechanics features included positive end-expiratory pressure (PEEP), respiratory rate (RR), peak inspiratory pressure, minute ventilation, tidal volume, mean airway pressure, and pressure-support level. Other cardiovascular and neurological features included heart rate, mean arterial pressure, temperature, and the Glasgow Coma Scale (GCS) total score.

Static features such as, patient age, duration of the ventilation episode, and the weight-normalized urine output rate. Demographics such as patient gender, and race have also been extracted but were held out for cohort auditing rather than downstream prediction, while patient weight was only extracted to derive the urine output rate normalized to body weight, and did not represent a predictive feature.

An initial feature set was limited to only variables required by the 2 Physiological Equations mentioned above, which ultimately proved to be insufficient for the downstream prediction task,

and the feature set has been expanded to cover additional clinical domains as the ones mentioned above. Several features from the analyzed past works have been explored but ultimately excluded from the feature set due to poor coverage in the real data. An example is Torrini et al.'s work, proposing negative inspiratory force [30], but has proven to have poor availability across the cohort with only 0.79% coverage. Arterial oxygen saturation (SaO_2) was also removed from the final downstream input set because of its poor availability, and available replacement (SpO_2) with significantly better coverage recommended by Dr. Murali.

The identified dynamic features were extracted from MIMIC-IV chart and laboratory event tables , which have then been matched to their corresponding ICU stay, and extubation events, then the arterial blood gas measurements were limited to samples from arterial specimen only. Furthermore, where instances of variables appear to have more than one measurement, the chosen solution was to take the mean of all the values. The GCS total was computed based on the latest valid eye, motor, and verbal components that were extracted with each episode interval, and once all 3 components were available.

5.4.2 Normalization, Unit Conversion, and Data Cleaning

Before conducting the temporal aggregation step, the measurements were normalized to a standard unit. For example, values of FiO_2 which were recorded as a percentage have been divided by 100. Additionally, other values were converted to a different standard unit, such as temperatures recorded in Fahrenheit degrees were ultimately converted into Celsius degrees. Then, plausible ranges were applied to each value in order to exclude clinically impossible measurements. For example, FiO_2 measurements were bounded between 0.20 and 1.00, SpO_2 between 0 and 100%, arterial pH between 6.0 and 8.5, and temperatures between 25 and 45°C. The GCS components of eye, motor, and verbal were also bounded by 1 to 4, 1 to 6, and 1 to 5 respectively, with the GCS total being bounded to values between 3 and 15. All other continuous feature measurements were only bounded to be positive, with the exception of PEEP and pressure support level where a value of 0 was allowed.

5.4.3 Temporal Representation of Episodes

For each extubation episode, only the measurements falling within the 6 hours decision window before the extubation event were extracted. The 6 hours decision window was then divided into 4 consecutive 1.5 hour bins ordered from the oldest interval to the youngest before the extubation event. This approach has created a more common temporal structure at the cost of varying clinical measurement coverage across the timestamps between patients.

The lookback mechanism used during the initial extraction experiment of only physiological variables related to the 2 equations has not been used to initialize measurements prior to the 6 hour decision window in the first bin. The main idea was to avoid introducing older measurements and propagating them into the decision window as some measurements may be outdated and irrelevant during that time.

5.4.4 Interpolation and Forward Filling

The missing values were initially addressed using values from the same extubation episode, where features such as, SpO_2, repiratory rate, mean arterial pressure, and temperature were linearly interpolated when entries were available on both bounds within the same episode. The interpolation technique was limited to filling missing measurements internally and did not extrapolate past the first or last recorded values avilable.

Another filling technique was that values could be propagated to as far as 3 subsequent bins, which is the equivalent of 4.5 hours. Such variables included FiO_2, $PaCO_2$, PaO_2, PEEP, peak inspiratory pressure, minute ventilation, tidal volume, mean airway pressure, arterial pH, GCS total score, and pressure support level. Hemoglobin is a feature rarely measured in clinical data,

so it was set to a forward filling window of up to 6 hours. Lastly, backward filling was not implemented, and missingnmess filling was conducted on an independent basis for each extubation episode, avoiding risks such as data leakage between patients or distinct extubation episodes.

Urine output rate was computed over the 24 hour window prior to the extubation event, where the interval included measurements prior to extubation and excluded the ones past. Urinary measurements were extracted from samples that included Foley catheter, voided urine, condom catheter, ileal conduit, supracubic catheter, nephrostomy, straight catheter, and ureteral stent. Other measurements, such as genitourinary irrigant input were converted to being negative before the sum so that total result represents the net urinary output. Furthermore, the extracted patient weight was used to normalize the output, where any weight below 20 kg was excluded. The normalized urine output rate [12] was computed according to the following formula:

$$\text{Urine Output Rate} = \frac{\text{Total Urine Volume (mL)}}{\text{Weight (kg)} \times \text{Time (hours)}} \tag{5.1}$$

Extubation episodes where patient weight measurements were missing also caused a missing urin output rate entry, but any other missing rates were handled by the multivariate imputation technique (MICE).

5.4.5 MICE Imputation

After the interpolation and forward filling techniques, the remaining missing measurements were imputed using the **miceforest** variation of Multivariate Imputation by Chaine Equations (MICE) [31]. The imputation matrix received all 4 bins in a wide representation in order to allow measurements from different timestamps before the extubation event to contribute. The static features have also been included in the MICE input, therefore producing a 75 column imputation matrix that included 18 dynamic and 3 static features across 4 consecutive timestamps before the extubation event.

The MICE imputation model parameters included LightBGM, 5 chained iterations, and predictive mean matching with 5 candidate values, generating a complete dataset. As a measure to prevent data leakage, the train-validation-test split was done before the imputation, so that the MICE model was fitted exclusively on the training split and was then applied to the validation and test splits without refitting.

5.4.6 Standardization and Masking

Before standardizing, features such as peak inspiratory pressure, minute ventilation, and mean airway pressure have been capped for higher bounds (upper-winsored) with 44 cmH_2O, 20 L/min, and 25 cmH_2O respectively to avoid definitive removal of upper tails. Then the dynamic features have been standardized using a standard scaler with the mean and standard deviation computed from real measurements within the training set. Essentially, for each dynamic feature, real recorded values across the 4 bins have been pooled and then used in computing the estimates mentioned. This method avoids training 4 separate standard scalers, one for each bin. The transformation was then applied to the validation, and test splits, as well as to the synthetic dataset, while the static features were simply copied over for each bin.

An important design choice was the implementation of additional binary columns indicating which values were really measured in the dataset and which ones were imputed. Therefore, each of the 17 dynamic features was followed by a column with a binary value stating whether it is a real measurement, and another binary value column indicating whether the feature value is forward filled, while also allowing for another similar column to indicate whether the value has been imputed using MICE if neither of the other 2 is positive. Lastly, another column also indicated whether the urine output rate is available, producing a total of 35 binary columns, which were

excluded from the standardization process.

Finally, the input sample at each bin included a total of 20 features (17 dynamic + 3 static) combined with 35 binary columns indicating real vs. filled vs. MICE imputation, totalling 55. Therefore, each extubation episode was represented by a tensor of 4 timestamps and 55 columns. Other data such as IDs, audit demographic features, and the extubation label were stored separately but not included in the downstream prediction input.

5.5 Synthetic EHR Data Generation

Synthetic extubation episodes were generated using a modified, off-the-shelf implementation of TimeDiff [13]. TimeDiff was selected as the model of choice due to its capability of joint representation of time-series clinical records and discrete variables. The model used in this project employs Gaussian Diffusion for the dynamic feature channels, and Multinomial Diffusion for the binary variables associated with the source of the corresponding feature value, as well as for the extubation label. Throughout the training process, the model has managed to learn the feature representation through a reversal process, and ultimately generate new data samples resembling the 4 step extubation episode structure.

To train the TimeDiff model, only a training split derived from real MIMIC-IV data within the extracted cohort was used, and neither validation nor any of the test samples were included to avoid leakage. The input tensor used for training had a shape of $(5{,}396, 56, 4)$, and consisted of 20 dynamic physiological features, 35 binary variables indicating the source of the corresponding feature value, and another binary variable representing the extubation label. To ensure all steps of an episode have the same label, the extubation label was copied across each of the 4 steps based on the corresponding extubation episode, in order to avoid any step within the same episode with a different label when generated. Essentially, the values corresponding t the dynamic features were estimated based on the real MIMIC-IV training samples, while the binary variables simply had a 0 or 1 representation.

The chosen TimeDiff model architecture consisted of 3 layers of LSTM, hidden size of 256, and 1,000 diffusion time steps. It was trained for 40,000 steps using parameters such as a batch of 32 and gradient accumulation across 2 batches, a learning rate of 0.00008, as well as a categorical loss weight of 0.8. For comparison purposes, training checkpoints were recorded at 5,000, 10,000, 20,000, and 40,000 training steps, where the generated samples used the weights from the 40,000 checkpoint. After the analysis conducted on the generated samples, which is discussed in more detail in Section (6.3), the TimeDiff model was then retrained with the newly derived upper winsorization implementations in the training data discussed in Section (5.4).

While the training process of TimeDiff did not impose any conditions to address the severe class imbalance, balanced rejection sampling was then employed to generate a balanced synthetic dataset where the underlying classes are balanced to 50-50. The synthetically generated episodes were separated based on outcome, and the process continued until each class accumulated 10,000 samples, creating a 20,000 samples synthetic dataset. This procedure did not alter any feature representations of the synthetic data nor involved retraining the model, but rather piled samples in each class until both reached 10,000 samples and rejected any extra ones in the mean time.

An important observation is that sometimes the generated data did not create samples where all 4 steps within an extubation episode corresponded to the same extubation label, so in order to address this, majority voting was employed and the winner was copied across each step. Then the synthetic dataset was further inspected for any missing or infinite values, non binary values assigned to the binary variables, and any episode label inconsistency.

In the context of predictive augmentation experiments, only the synthtetically generated extubation failure episodes were utilized, while the real training set derived from MIMIC-IV consisted of 4,844 successful extubation labels and 552 extubation failure labels, defining a discrepancy of 4,292 extubation failure episodes. To address the deficit, a subset of 4,292 synthetically generated

extubation failure episodes were added on top of the real ones in the training partition to balance the class distribution through augmentation while retaining as much real records as possible. Therefore, the newly augmented training set consisted of 4,844 extubation failure episodes and 4,844 extinaton success episodes, and included another binary variable identifying the synthetically generated samples from the real, forward filled and interpolated, and MICE imputed ones, but was purposefully excluded when conducting this experiment to retain the original 55 channel feature structure.

Lastly, the real data validation and test splits derived from MIMIC-IV remained unchanged throughout the augmentation experiment in order to properly assess whether the synthetic data can provide the same feature representations for the model when evaluated on unseen real data. Furthermore, no preprocessing parameters were estimated again using the synthetically generated samples, with any numerical transformation remaining solely dependent on the real patient records. The evaluation procedure aimed to check whether adding additional synthetically generated samples to compensate poor class distribution of the minority class could improve predictive capability on unseen real measurements. A more in depth analysis of statistical fidelity of the synthetically generated episodes, as well as the underlying results of the experiments can be found in Sections (6.3 and 6.4).

5.6 Predictive Modelling

The predictive downstream task was defined as a binary classification at extubation episode level, meaning that for each extubation episode, the model receives 4 steps of consecutive 90 minutes intervals based on a 6 hour decision window, and 55 corresponding features, while returning a probablity of extubation failure. A breakdown of the 55 input features are described in more detail in Section (5.4). All models were only trained using the designated training split derived from MIMIC-IV, while the validation split was utilized for hyperparameter selection, early stopping, and decision threshold selection. Furthermore, the test split was held out for the final evaluation on unseen data.

The XGBoost architecture was selected as a tree based baseline model [32], because boosted decision trees do not process time-series tensors, such that the the 4 step extubation episodes were treated as separate fixed length feature vectors. Furthermore, a total of 4 XGBoost model configurations were evaluated, where the first one consisted solely of 6 physiological features that are required to compute the 2 physiological equations, a second configuration that uses an expanded feature space of 20 dynamic and static physiological features but excluded the binary variables indicating source of the corresponding value, a third configuration that incorporates the 35 binary variables to account for feature source totalling 55 features, and a fourth configuration that incorporates the 55 features as well as positive class weighting to address class imnbalance within the MIMIC-IV cohort. The evaluation procedure of these configurations aimed to assess whether an expanded feature space, feature value source information, and explicit weight handling of the minority class contributed to the predictive performance of the model.

Fine tuning the XGBoost model explored various learning rates of 0.01, 0.02, and 0.05, a maximum tree depth of 2, 3, or 4, L2 regularization term of 1, 5, or 10, an L1 regularization term of 0, 0.1, or 1, a minimum child weight of 5, 10, or 20, and a sumbsampling proportion of 0.7, 0.85, or 1. Additionally, boosting rounds maximum was set to 2,000, and training stopped after 50 epochs without significant improvement in the validation AUPRC score. Other parameters include a random seed set to 42 to ensure experiment reproduction, and CUDA was utilized to speed up the training process. Lastly, the weight required to balance the class distribution was computed based on the training set samples only.

On the other hand, fine tuning GRU and LSTM networks [33], [34] were explored as sequential architecture baselines. Unlike the XGBoost architecture, GRU and LSTM are capable of processing the 4 step time-series steps directly, where each recurrent network would receive an input with feature space 55 for each of the 4 steps across all extubation episodes, while the extubation label is represented as a binary variable. Threfore, the final output label represents the predicted

probability of extubation failure for a complete extubation episode compared to a separate label for each time step.

Training of the GRU and LSTM models involved using a batch size of 64, a maximum epoch number set to 50, early stopping was set to freeze the weights after 10 consecutive epochs do not exhibit improvement, while the version with the highest validation AUPRC score was kept. Various configurations of the GRU and LSTM models were evaluated, such as a baseline GRU model consisting of one recurrent layer, hidden size of 32, no dropout percentage, a learning rate of 0.001, and a positive class weight of 8.775, with the best checkpoint selected at epoch 12. Additionally, the chosen LSTM model consisted of 2 recurrent layers, hidden size of 64, a dropout of 20%, learning rate of 0.001, and a positive class weight of 4.0, with the best checkpoint selected at epoch 41.

Demographic features such as patient gender and race were not included in the primary feature space of the downstream prediction task. Further sensitivity analysis was conducted on both demographic features one hot encoding them and feeding them as inputs to the GRU and LSTM models before training. The evaluation of this experiment was based on validation AUPRC score and compared to the models with similar configurations but lacking the demographic features from the feature speace. These demographic features were ultimately excluded from the feature space due to no improvement in the predictive capability of the model, and were therefore retained for cohort description and subgroup analysis.

Additional exploratory analysis was conducted for the GRU and LSTM models using the synthetically generated extubation episodes discussed in Section 5.5 for data augmentation. Fine tuning under this context consisted of hyperparameters such as hidden size of 32, or 64, a single or 2 recurrent layers within the architecture, dropout values of 0 or 0.2, learning rates of 0.0003 or 0.001, batch size of 64, maximum training epochs of 50, and early stopping limimted to 10 consecutive epochs with no improvement. Additionally, no class weighting was implemented for the minority class because data augmentation is aimed to address class distribution imbalance in this scenario, with the augmented dataset containing a 50-50 split of extubation failure and successful extubation episodes. The models were evaluated on the real validation set derived from MIMIC-IV.

The GRU and LSTM models trained solely on real measurements discussed in more detail in Section (6.4) were selected as the pre-augmentation baselines, while model baselines trained during the physics-informed experiment were trained and implemented as a separate comparison benchmark, which is discussed in more detail in Section (5.7). By employing this approach of training separate models during distinct experiments, we can ensure that the physiological constraints are assessed against models trained within similar experimental means rather than old checkpoints.

5.7 Physics-Informed Component

The physics-informed model was constructed based on the GRU architecture amid the high validation AUPRC scores achieved by GRU among to other recurrent baselines. This model received as input the same 4 time step, 55 channel tensor, which has been discussed in more detail in Section (5.6). This GRU model was connected to 2 separate output branches, where the first branch produced the extubation failure probability on an episode level, while the second branch produced 7 physiological values at each of the 4 time steps, which consisted of FiO_2, $PaCO_2$, PaO_2, SpO_2, Hemoglobin, alveolar oxygen pressure, and arterial oxygen content. Throughout this section, the term physiological state will represent this set of 7 auxiliary physiological outputs rather than an internal hidden state within the GRU model architecture.

The first 5 auxiliary outputs correspond to measurements that are available in the patient records, while the remaining 2 represent quantities derived from the physiological equations. That being said, the alveolar oxygen pressure was computed using the Alveolar Gas Equation presented in Equation (4.1). In the implemented constraint, the atmospheric pressure was set to a fixed value of 760 mmHg, water vapor pressure was set to 47 mmHg, and the respiratory quotient to 0.8. Furthermore, the Arterial Oxygen Content Equation was implemented using Equation (4.2). In the implemented constraint, SpO_2 was substituted for SaO_2 because SpO_2 was the oxygen

saturation variable with higher real data availability, and the dissolved oxygen coefficient was set to 0.0031. The resulting implementation of arterial oxygen content was expressed in O_2/dL.

In order to link the auxiliary outputs to the available patient record values, a state prediction loss was implemented to address the matter. This loss aimed to penalize apply a penalty for differences between values predicted by the auxiliary output branch and the corresponding target, while applying a mask to ensure updates are only applied if the measurements were available at that time step. On the contrary, this model did not apply penalties for value sources other than real data, meaning that missed values originating from filling or MICE imputation did not receive a penalty. Targets for the alveolar oxygen pressure were constructed if and only if values for features such as FiO_2 and $PaCO_2$ were available. Additionally, targets for the arterial oxygen content also required available values for features such as Hemoglobin, SpO_2, and PaO_2.

The 2 physiological equations were implemented as soft constraints rather than replacing the observed feature space. That is, for each episode time step, an alveolar gas residual estimates the difference between the predicted alveolar oxygen pressure and the value obtained when computing Equation (4.1) using the other auxiliary outputs. Similarly, the oxygen content residual served a similar purpose using Equation (4.2). By applying a penalty to these residuals, the auxiliary outputs are forced to maintain mutual consistency with the 2 physiological equations, while also allowing the outcome branch to learn representations from the directly measured clinical values.

In equation form, the training objective of this implementations is the following:

$$\mathcal{L}\text{outcome} + \alpha\text{state}\mathcal{L}\text{state} + \beta\text{alveolar}\mathcal{L}\text{alveolar} + \beta\text{oxygen}\mathcal{L}\text{oxygen} \tag{5.2}$$

Where, $\mathcal{L}$outcome represents the binary extubation label classification loss, $\mathcal{L}$state is the masked state prediction loss, while $\mathcal{L}$alveolar and $\mathcal{L}$oxygen represent the penalized terms based on the 2 physiological equations. Furthermore, the α and β coefficients controlled the contribution degree of each auxiliary term relative to the extubation outcome prediction. After an exploratory analysis, the final configuration of this component used $\alpha_{\text{state}} = 0.01$, $\beta_{\text{alveolar}} = 0.05$ and $\beta_{\text{oxygen}} = 0.01$ amid achieving high AUPRC validation scores.

A series of 6 experiments were conducted to identify the contributions of each loss component, with all experiments maintaining the same extubation outcome prediction downstream task and class weightings to ensure fairness. The 6 experiments consisted of a baseline GRU model using only an outcome loss, a second model that incorporated the state prediction loss, another model incorporating just the Alveolar Gas Equation loss, as well as a model incorporating just the Oxygen Content Equation loss, one model implementing both physiological equation losses but excluding the state prediction loss, and a sixth model combining the extubation outcome, state prediction, and the 2 physiological equation loss terms. The model incorporating both equation losses but omitting the state prediction loss could produce outputs satisfying the 2 physiological constraints without requiring to link the outputs to any patient records.

Each of the 6 models was trained on the real training split derived from MIMIC-IV, exlcuding any synthetic data augmentation from these experiments. To ensure fairness throughout and better assess loss term impact, the model architectures, input representations, data splits, and training hyperparameters were kept constant. The best checkpoints were selected based on validation AUPRC scores, while noting that a separate decision threshold was derived for each model using the validation data split, which were ultimately frozen and unchanged when evaluated on the held-out test split. An analysis was conducted by comparing the models' predictive performance on the test split, as well as the residual errors of both physiological equations. A more in depth analysis of the experiments and the results can be found in Section (6.5)

5.8 Experimental Design and Evaluation Metrics

All predictive experiments consisted of the same train, validation, and held-out test splits derived from the MIMIC-IV cohort, as described in Section (5.6). In order to ensure that an episode does not appear in more than one data split, the data was divided based on patient rather than extubation episodes, so that all episodes belonging to any one patient would be partitioned in the same split. The training set was utilized to fit the models' parameters, while the validation set was primarily used for best hyperparameter and decision threshold identification, and the held-out test split being reserved for pure evaluation with all parameters frozen. The extubation failure outcome was treated as the positive class throughout the entire deveopment and evaluation process.

The area under the precision recall curve (AUPRC) was selected as the primary metric for model selection because of a severe class imbalance where extubation failure outcomes represented approximately 10% of the entire cohort. AUPRC measures the performance of a model's ability to identify the positive class based on a balance between precision and recall. Furthermore, the area under the receiver operating characteristic curve (AUROC) was selected as a complementary metric for threshold discrimination.

Other metrics such as precision, recall, F1 score, and true negative rate (specificity) were used to describe model behavior and assess performance at various thresholds. Precision measured the what proportion of the predicted extubation failures were true failures, while recall measured the proportion of extubation failures correctly identified by the model. Furthermore, the true negative rate measured what proportion of successful extubations were correctly identified, and the F1 score produced the harmonic mean of precision and recall. Confusion matrix counts of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) were also reported when evaluating model performance.

The 3 XGBoost configurations lacking positive class weighting were all evaluated using a default probability threshold of 0.5, while for the class weighted XGBoost, GRU, and LSTM architectures the probability threshold was derived by maximizing the F1 score on the validation split. Decision thresholds belonging to the physics informed models were derived in a similar manner. Lastly, once a threshold is derived, it is frozen and applied unchanged to the held-out test set, while neither the model configuration, hyperparameters, nor decision threshold was adjusted amid the results obtained during evaluation on the held-out test split.

A fidelity assessment was conducted on the synthetically generated databy comparing the generated episodes with the real measurements from the training split utilized to train the TimeDiff generative model. The Kolmogorov-Smirnov statistic was computed on a feature basis aimed to compare the shapes of real and synthetic marginal distributions. The Wasserstein Distance was also computed and divided by the interquartile range on a feature basis to measure the distances between synthetic and the corresponding real feature values, while also allowing variables with distinct measurement scales to be compared and assessed. The Komlogorov-Smirnov statistic and Wasserstein Distance results should be interpreted in the sense that a lower value indicates closer agreement between the values. Furthermore, the median and 99th percentile values were examined to identify the centers and upper bounds of the distributions, with the synthetically generated values that fell outside the real data range being used to identify extremes that were clinically not plausible. Lastly, positive versus negative median differences were compared to determine if class relationship directions that exist in the real data were mirrored in the synthetic samples.

In order to assess the degree of contribution of synthetic data augmentation on the downstream prediction task, GRU and LSTM models were trained using the train split alone, leaving the real validation and test splits untouched. The augmented models were ranked based on their AUPRC score on the validation split, and then compared with models of the same configuration but trained on real data only. Synthetically genreated extubation episodes were incorporated in the training procedure only, such that no direct influence could be transmitted into the validation and test metrics.

The models developed during the physics-informed experiments were evaluated by assessing

both predictive performance of extubation outcome, as well as physiological consistency with the physiological constraints. The predictive performance was measured using metrics such as AUPRC, AUROC, and the other metrics that are dependent on the decision threshold to establish the baseline, while physiological consistency was evaluated based on the mean absolute residual of each of the 2 physiological equations on the held-out test split. A lower residual value indicates better agreement with the corresponding equation, but was not considered relevant when evaluating extubation outcome prediction performance. Furthermore, the alveolar gas equation residual was measured in mmHg, while the arterial oxygen content residual was measured in mL O_2/dL.

An exploratory analysis experiment of 2,000 bootstrap resamples at patient level of the held-out test set was used to estimate the uncertainty in the difference between the complete PINN model and the underlying GRU baseline model. Patient sampling was done using replacement, and all extubation episodes corresponding to a specific patient were grouped together. Both the full PINN model and the baseline were evaluated on the same set of resampled patients with the difference estimated between their metrics was recorded at each repetition. The resulted distributions were then used to compute the 95% confidence intervals for differences in metrics such as precision, recall, F1 score, true negative rate, AUROC, and AUPRC, while noting that neither model was subject to retraining during this experiment.

Lastly, external validation was conducted using eICU-CRD after the eligible episodes were transformed into the same four-time-step, 55-channel input representation used for the MIMIC-IV cohort. The selected full PINN model described in more detail in Section (5.7) was applied using its unchanged weights and parameters derived using the MIMIC-IV validation set. The model was not retrained, and the decision threshold of 0.5892 derived from MIMIC-IV stayed the same with no further recalibration for eICU-CRD. Predictive performance was assessed using metrics such as AUROC, AUPRC, precision, recall, true negative rate, F1 score and the confusion matrix counts of TP, TN, FP, and FN. Additionally, the corresponding 95% confidence intervals were estimated using 2,000 bootstrap resamples at episode level. The purpose of this analysis was to analyze how transferable the frozen model would be to a distinct critical care dataset.

Chapter 6

Results and Evaluation

6.1 Cohort and Dataset

Based on the construction procedures of the primary cohort outlined in Section (5.2), a total of 7,713 valid extubation episodes were extracted. After further preprocessing and filtering, specifically by implementing the 30 day upper bound to the ventilation duration, 7 extubation episodes have been removed from the cohort, which represented approximately 0.09% of the extracted cohort. From that, the resulting cohort consisted of 7,706 valid extubation episodes based on 7,085 unique patients.

6,922 episodes were labeled as successful extubation, and the remaining 784 as extubation failure, where each accounted for approximately 89.83%, and 10.17% respectively. From the 784 extubation episodes labeled as extubation failure, 735 satisfied the reintubation within 48 hours of extubation criterion, while the remaining 49 satisfied the prolonged non invasive respiratory support for longer than 48 hours after the extubation event criterion. Furhtermore, according to the outcome distribution, extubation failure is undenyably a minor class in the extracted cohort dataset. A more detailed summary of the characteristics of the final MIMIC-IV cohort based on extubation outcome can be found in Table (6.1).

In terms of demographic characteristics, both extubation outcomes presented relatively similar distributions. The median patient age was 65 and 66 for successful extubation and extubation failure respectively, with the interquartile range (IQR) being 54 to 75 years in both classes. Genderwise, approximately 58.0% of successful extubation patients were males, while 58.4% of patients where extubation has failed were also males. Furthermore, in terms of race, white patients made up the majority of the extracted cohort with approximately 61.5% of all extubation episodes, while the second largest majority was populated by patients' whose race or ethnicity was not known, accounting for 17.7%.

Duration of invasive ventilation prior to the extubation event defined a clear difference between the 2 classes, with the median ventilation duration of 91.7 hours for extubation failure and 69.5 hours for patients who have been successfully extubated. The ventilation duration IQRs were 41.6 to 137.3 hours and 50.2 to 163.1 hours for extubation success and extubation failure respectively. However, the median urine output rate normalized by patient weight was 1.0 across the entire dataset, representing a feature that does not carry much weight in the exubation outcome prediction, which is a contrary finding to past works such as Zhao et al. [12] and Pai et al [29].

After splitting each extubation episode into 4 consecutive, 90 minute steps, a total of 30,824 entries were obtained, and during the cleaning stage no duplicates of either episodes or steps were identified. Therefore, all 30,824 records were valid, where each extubation episode was represented by exactly 4 consecutive steps, and the extubation outcome did not differ in any of the steps for any episode.

Lastly, the dataset was divided at a patient level into training, validation, and test splits of 70%, 15%, and 15% respectively. The resulting splits consisted of 5,396 training samples, 1,144

validation samples, and 1,166 test samples. Further validation checks were conducted to ensure that no patient appears in more than one split, and that measurements from separate extubation episodes belonging to the same patient also did not appear in more than one split.

Table 6.1: Characteristics of the final MIMIC-IV cohort based on the extubation outcome

Characteristic	**Cohort**	**Extubation Success**	**Extubation Failure**
Episodes, n	7,706	6,922	784
Patients, n	7,085	N/A	N/A
Age (years), median (IQR)	65.0 (54.0-75.0)	65.0 (54.0-75.0)	66.0 (54.0-75.0)
Female, n (%)	3,230 (41.9%)	2,904 (42.0%)	326 (41.6%)
Male, n (%)	4,476 (58.1%)	4,018 (58.0%)	458 (58.4%)
Race/ethnicity, n (%)			
White	4,736 (61.5%)	4,252 (61.4%)	484 (61.7%)
Black/African American	765 (9.9%)	707 (10.2%)	58 (7.4%)
Hispanic/Latino	261 (3.4%)	232 (3.4%)	29 (3.7%)
Asian	212 (2.8%)	188 (2.7%)	24 (3.1%)
Other	365 (4.7%)	322 (4.7%)	43 (5.5%)
Unknown	1,367 (17.7%)	1,221 (17.6%)	146 (18.6%)
Ventilation duration (hours), median (IQR)	71.7 (41.6-137.3)	69.5 (41.1-134.7)	91.7 (50.2-163.1)
Urine-output rate (mL/kg/h), median (IQR)	1.0 (0.6-1.5)	1.0 (0.6-1.5)	1.0 (0.6-1.6)
Reintubation within 48 hours, n (%)	735 (9.5%)	0 (0.0%)	735 (93.8%)
Prolonged non-invasive respiratory support ($>$ 48 hours), n (%)	49 (0.6%)	0 (0.0%)	49 (6.2%)

This table summarizes the characteristics distribution with dynamic feature values provided in median IQR, and categorical values in their total number counts and resulting percentage. This all corresponds to the extubation failure label.

6.2 Dataset Suitability

The Dataset Suitability Assessment procedure was conducted against the primary 20 features selected for the downstream prediction task, meaning that the ability to extract the 20 features, reconstruct the two extubation failure conditions, and compatibility with the 6 hour decision window in temporal representation are crucial to accomplish this project's goals. The datasets compared in this procedure consisted of MIMIC-IV, eICU-CRD, and the Royal Brompton Hospital (RBH) dataset. Of the 3, MIMIC-IV and eICU-CRD contained and supported the computation of all the 20 features required for the downstream task. On the other hand, the Royal Brompton Hospital dataset only supported 18 of the 20 features, with features such as temperature and numerical pressure-support level being unavailable. A more detailed per-feature availability comparison across the 3 datasets can be observed in Table (6.2).

A dataset that has not been specified above but was initially considered is AmsterdamUMCdb. While being briefly explored in the early stages of the project, it was ultiamtely exluded from a complete suitability analysis due to a large number of features, specifically categorical and labels, containing a large amount of entries in Dutch. For example, several gender entries consisted of *Vrouw*, which translates to *Female*, as well as clinical specialty labels that included *Vaatchirurgie* and *Heelkunde Gastro-enterologie*, which ultimately translate to *Vascular Surgery* and *Gastrointestinal Surgery*. While a language translation library, such as *deep-translate* could have been used to normalize the clinical terminology and convert the Dutch entries into consistent English ones, significant overhead would have been added to the already complex project. Full translation of the dataset would have required not only conversion between Dutch and English, but also implied risk

of mistranslating misspellings, deeper verification and validation of the clinical context meaning, all being necessary before being able to reliably begin the dataset suitability analysis. The implementation overhead would have significantly impacted the task implementation schedule of the already complex project due to a lack of concrete evidence that AmsterdamUMCdb met the model development or validation requirements. Therefore, the exclusion of this dataset was a decision taken based on project feasibility given the timeline rather than a direct proof that the dataset is unsuitable for the project's downstream task.

Table 6.2: Feature Availability Comparison across MIMIC-IV, eICU-CRD, and Royal Brompton Hospital Dataset.

Feature	**MIMIC-IV**	**eICU-CRD**	**RBH**
Dynamic Features			
FiO_2	Yes	Yes	Yes
$PaCO_2$	Yes	Yes	Yes
PaO_2	Yes	Yes	Yes
SpO_2	Yes	Yes	Yes
Hemoglobin	Yes	Yes	Yes
PEEP	Yes	Yes	Yes
Respiratory rate	Yes	Yes	Yes
Heart rate	Yes	Yes	Yes
Peak inspiratory pressure	Yes	Yes	Yes
Minute ventilation	Yes	Yes	Yes
Tidal volume	Yes	Yes	Yes
Mean arterial pressure	Yes	Yes	Yes
Mean airway pressure	Yes	Yes	Yes
Arterial pH	Yes	Yes	Yes
Temperature	Yes	Yes	**No**
GCS total	Yes	Yes	Yes
Pressure-support level	Yes	Yes	**No**
Static Features			
Age	Yes	Yes	Yes
Ventilation duration	Yes	Yes	Yes
Urine-output rate	Yes	Yes	Yes

This table presents feature availability comparison across the 3 selected datasets, consisting of MIMIC-IV, eICU-CRD, and Royal Brompton Hospital. A value of "Yes" indicates feature availability, while "No" indicates unavailability.

While the Royal Brompton Hospital dataset contained a feature called *support_mode*, but it is categorical and could not be considered a direct equivalent to the numerical pressure-support level value required by the downstream task. By removing either *temperature* or *pressure-support level* would have impacted the model's feature specification and require retraining, as well as re-evaluation on new datasets excluding the 2 underlying features. Therefore, the Royal Brompton Hospital dataset was retained as an additional component for the dataset suitability assessment rather than being utilized for external validation purposes. By comparison, the eICU-CRD dataset supported the full feature set and outcome condition reconstruction capability, resulting in a final external validation cohort consisting of 13,595 extubation episodes from 12,590 unique patients. However, a downside of this dataset was also the lack in positive samples with a class split of approximately 6.69% and 93.31% for positive and negative labels respectively. The external validation results are reported in Section (6.6).

In the MIMIC-IV dataset, feature coverage at episode level of features such as *temperature*, *GCS total*, and *pressure-support level* presented strong results of 93.29%, 92.20%, and 85.62% respectively. Other features attempted to include during the feature expansion procedure included *negative inspiratory force*, which was ultimately excluded due to poor coverage of 0.79%, which means that only 61 episodes contained this metric. *Patient age* and *ventilation duration* presented

an episode coverage of 100%, and were therefore retained in the feature set. Lastly, *urine-output rate* has also proven to be a strongly covered feature with approximately 73.90% before the imputation procedure, accounting for approximately 5,695 extubation episodes, with only 2,011 samples requiring imputation.

The coverage of dynamic features varied significantly, where more routinely recorded variables such as heart rate, respiratory rate, mean arterial pressure, and SpO_2 presented coverage of approximately 98%, while other variables such as FiO_2 was covered approximately 43% by direct recordings, 34% by forward filled and interpolated values, and the remaining 23% by MICE imputation. Another dynamic feature that showed a similar pattern is PEEP, which was covered approximately in proportions of 40%, 39%, and 21% by direct measurements, forward filling and interpolation, and MICE imputation. This discrepancy represents feature priority during intensive care monitoring during the final 6 hours prior to an extubation attempt. The forward filling and interpolation stage also contributed substantially to ventilator settings features such as minute ventilation, mean airway pressure, tidal volume, and peak inspiratory rate indicating that measurements were conducted periodically yet consistent enough to perform up to 4.5 hours interval of forward filling. The pressure-support level feature indicated less direct measurements in the 6 decision window with an almost equal split between measured, filled, and MICE imputed values.

Laboratory features presented high dependency on MICE imputation, which consisted of approximately 75% of all values for $PaCO_2$, PaO_2, and arterial pH, while Hemoglobin presented the highest level of dependency with approximately 83% of the samples being imputed using MICE. This dependency makes clinical sense as arterial blood gas and laboratory tests are recorded more sporadically on a need-basis based on the patient's status. Features such as temperature and GCS total were both covered approximately 40% by direct measurements, with temperature being more reliant on multivariate imputation, which accounted for roughly 35% of all samples. Overall, these results support the design choice of retaining the source indicators for each feature, because the physiological component was not fully measured by medical professionals, and would further enforce explainability if the model is able to differentiate between direct measurements, forward filled and interpolated values, and multivariate imputations, therefore creating a clear separation between real and preprocessed data. Figure (6.1) presents a visual representation of the feature source distribution in MIMIC-IV.

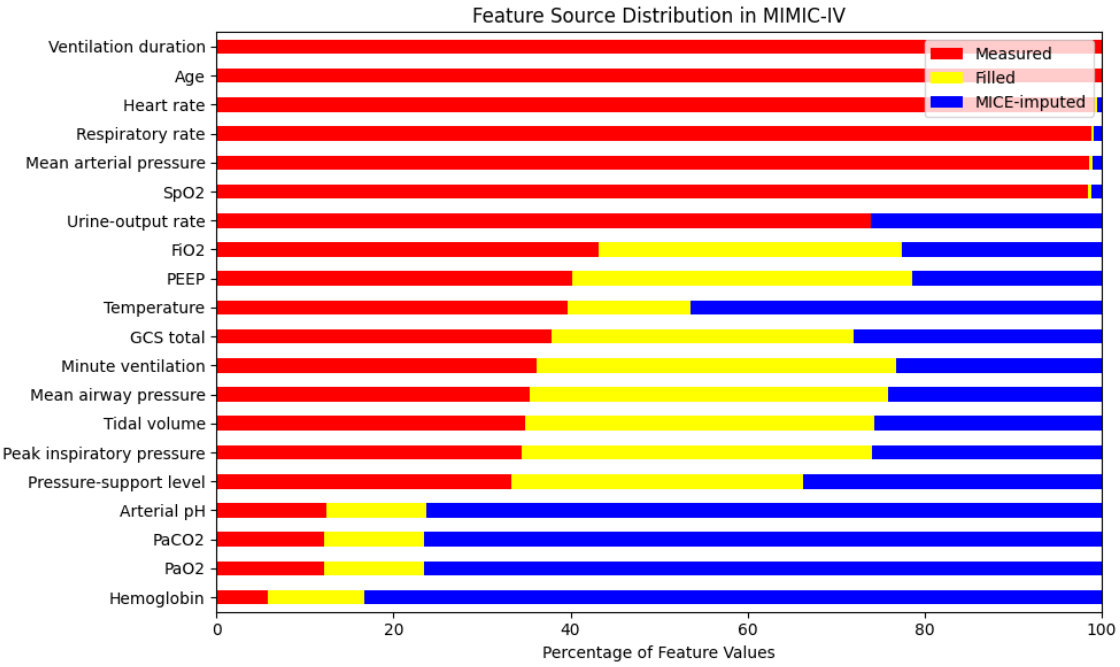


Figure 6.1: This plot presents a visual representation of the feature source distribution in MIMIC-IV, where red represents the values measured at collection time by the medical professionals, yellow represents the values filled through forward filling and interpolation, and blue represents the values imputed using the MICE method.

In the eICU-CRD dataset, feature coverage at episode level was significantly different from MIMIC-IV. Features such as patient age and ventilation duration were, similarly with MIMIC-IV, available entirely through clinical measurements, with heart rate and SpO_2 also requiring minimal filling and multivariate imputation. Respiratory rate was more reliant on MICE imputation in eICU than MIMIC, but still consisted of approximately 89% of samples originating from direct recordings of medical professionals. Similarly, urine-output rate consisted of approximately 62% of real measurements and 38% MICE imputed values, which is only slightly more dependent on imputation than in MIMIC. The largest differences could be observed in the respiratory and ventilator settings features, where approximately 35% of FiO_2 values were measured, 20% forward filled and interpolated, and 45% imputed. Features such as mean arterial pressure, GCS total score, and temperature displayed high dependency on MICE imputation, where temperature values consisted of approximately 83% multivariate imputations. Ventilator settings features also heavily relied on multivariate imputation, where PEEP had the lowest percentage of imputed values at approximately 60%, while peak inspiratory pressure consisted of roughly 80% imputed entries. Lastly, PaO_2, $PaCO_2$, arterial pH, and Hemoglobin were the least directly recorded features with less than 10% and more than 82% imputations each. Figure (6.2) presents a visual representation of the feature source distribution in eICU-CRD.

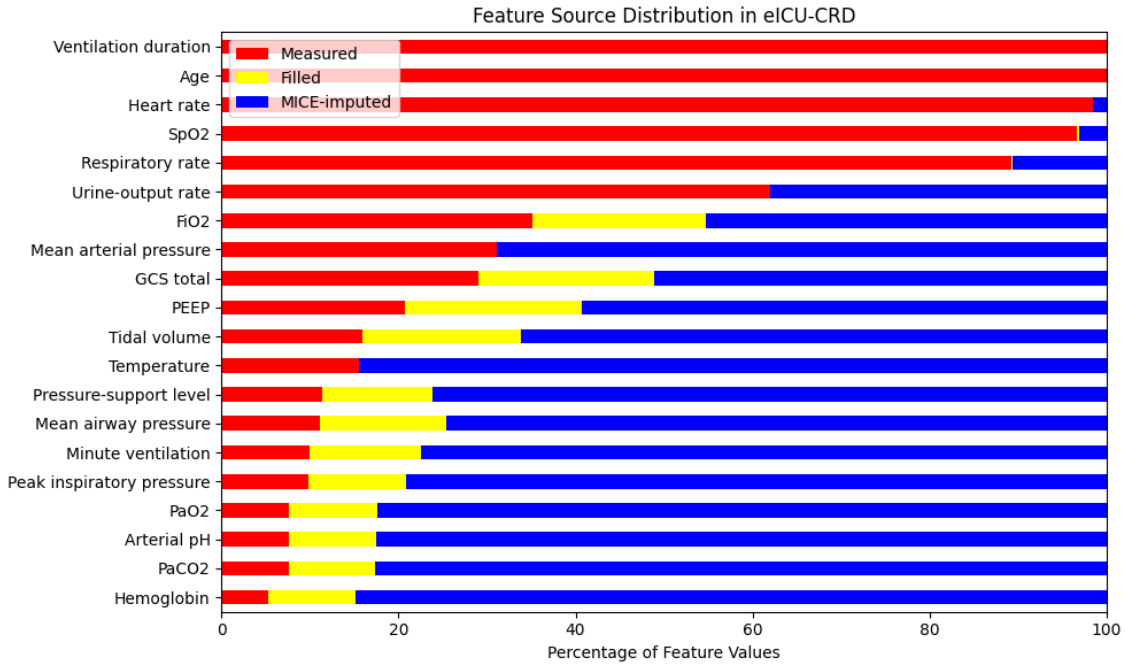


Figure 6.2: This plot presents a visual representation of the feature source distribution in eICU-CRD, where red represents the values measured at collection time by the medical professionals, yellow represents the values filled through forward filling and interpolation, and blue represents the values imputed using the MICE method.

In conclusion, both Figures (6.1 and 6.2) indicate that simply containing a feature does not imply automatic task suitability because frequency of direct measurements may differ, sometimes dramatically, between datasets. This is clinically plausible because even though the 2 datasets both originate from the United States, they differ from many perspectives, such as collection strategy, hospital system, collection timeline, and total patients just to name a few, with a more detailed analysis of each dataset done in Section (4.4). Despite eICU-CRD consisting of all the required features, it still relied significantly more heavily on MICE imputation than MIMIC-IV for features such as mean arterial pressure, temperature, GCS total score, and the ventilator settings variables. Arterial blood gas features and hemoglobin were poorly covered by direct measurements in both datasets in a similar amount, therefore meaning that the model would learn a similar feature representation from either dataset. Based on this analysis, MIMIC-IV has been selected as the primary development and internal evaluation dataset, while eICU-CRD was held out and utilized in the external evaluation process. To maintain an unbiased external validation procedure, the missing values in eICU-CRD requiring imputation were subjected to the same MICE model that was trained on the MIMIC-IV data and retained the binary features indicating whether a value is directly measured, forward filled and interpolated, or MICE imputed. Despite these design choices, real data sparsity and high dependency on multivariate imputation has caused a data shift, which has further negatively impacted the predictive capability of the PINN model during the external validation stage.

6.3 Synthetic EHR Data

An off-the-shelf TimeDiff model was trained using the MIMIC-IV training split, which consisted of a total of 5,396 extubation episodes. However, before the training data was suitable to be inputted into the TimeDiff model, it was necessary to convert it to a TimeDiff-specific tensor, which had to be divisible by 8, based on the available documentation and channel structure of the model, but that would have produced poor feature coverage at the step level. A design choice of 4 channels was selected with the risk of losing some degree of generalizability, but gaining more

real measurement coverage to benefit forward filling and interpolation, as well as diminishing the need for imputation by switching from 8 steps of 45 minutes each to 4 steps of 90 minutes each. The resulting TimeDiff training tensor had a shape of (5,396, 56, 4), consisting of 20 channels that represent the required features, as presented in Section (6.2), 35 channels of binary variables indicating whether the feature value is a direct measurement, forward filled and interpolated, or multivariate imputed, as well as the 4 steps representing 4 equal 90 minute sequences of the extubation episode. The architecture utilizes Gaussian Diffusion for the 20 required features, and Multinomial Diffusion for the 35 binary variables as well as for the extubation label. The model was trained for a total of 40,000 steps with checkpoints saved at 5,000, 10,000, and 20,000 steps, with the synthetic samples produced by the 40,000 step version being the ones utilized in the model evaluation.

The first batch of synthetic samples generated by the TimeDiff model reproduced central values of several features relatively well, but lacked capability to generate upper bound values for some of the ventilator settings features, and dramatically exceeded for others. For example, features such as peak inspiratory pressure presented a directly measured value in the 99th percentile of 34.0 $cmH2O$, while the synthetically generated 99th percentile value was 214.691 $cmH2O$. Similarly, minute ventilation presented a 99th percentile value of 16.2 L/min and a synthetically generated one of 76.160 L/min. Further inspection has also identified extreme upper bound values for various features, such as peak inspiratory pressure, minute ventilation, and mean airway pressure of 2,523 $cmH2O$, 777 L/min, and 96 $cmH2O$. In order to limit the influence of these extreme outliers without removing the corresponding extubation episodes, because they may represent severely ill patients, an upper bound clipping, or upper winzorized method was implemented exclusively for the 3 features mentioned. Therefore, peak inspiratory pressure was upper capped at 44 $cmH2O$, minute ventilation at 20 L/min, and mean airway pressure at 25 $cmH2O$. This implementation resulted in impacting 17, 16, and 13 samples respectively, and then the TimeDiff model was retrained on the same training split with the upper winsorization cap.

Table 6.3: Tail clipping based on training data applied before TimeDiff training.

Feature (unit)	**Upper Percentile**	**Upper Bound**	**Original maximum**	**Total Values Capped**
Peak inspiratory pressure (cmH_2O)	99.9	44.0	2,523	17
Minute ventilation (L/min)	99.9	20.0	777	16
Mean airway pressure (cmH_2O)	99.9	25.0	96	13

This table presents the upper winsorization cap implementation before retraining the TimeDiff model.

After retraining the model on the new configuration, it produced significantly more plausible distributions for peak inspiratory pressure, minute ventilation, and mean airway pressure. The synthetically generated peak inspiratory pressure median decreased from 32.368 to 13.609 $cmH2O$, compared to the directly measured median of 12.0 $cmH2O$, and similarly, the 99th percentile value has decreased from 214.691 to 26.163 $cmH2O$. Furthermore, the synthetically generated minute ventilation median has decreased from 13.159 to 8.631 L/min, which is nearing the directly measured median of 8.3 L/min, while its 99th percentile value has also decreased from 76.160 to 13.853 L/min. Lastly, mean airway pressure presented the least dramatic change with the synthetically generated median dropping from 7.780 to 7.529 $cmH2O$ nearing the real data median value of 7.0 $cmH2O$.

Further analysis has also identified that upper winsorization improved overall feature statistics, with the mean Kolmogorov-Smirnov statistic across features dropping from 0.194 to 0.156, which accounts for a reduction of approximately 19.6%. Additionally, the median Wasserstein Distance normalized to the interquartile range (IQR) of directly measured values has also decreased from 14.744% to 14.068%, which ultimately accounts for a decrease of 4.6%. Lastly, and most importantly, none of the produced values fell outside the corresponding lower and upper bounds set by the respective directly measured features.

Table 6.4: Statistical fidelity of synthetic data before and after value clipping to the upper bound.

Training Split	**Mean KS Statistic**	**Median of Normalized Wasserstein Distance (%)**	**Values Outside Real Data Range (%)**
Before Clipping	0.194	14.744%	N/A
After Clipping	0.156	14.068%	0.000%

This table presents the statistical fidelity of synthetic data before and after upper winsorization.

Other features were also affected by similar upper tail differences, which improved after retraining the TimeDiff model with upper winzoring. For example, the synthetically generated 99th percentiles were below the directly observed ones, with FiO_2 measuring 0.614 compared to 1.00, PEEP with 9.67 compared to 12.00, heart rate with 118.545 compared to 131.000, pressure-support level with 11.205 versus 15.00, tidal volume with 814.060 compared to 929.00, and ventilation duration with 274.453 compared to 423.421 hours, which was also identified as the feature with the least improvement. Additionally, while some feature representations have improved dramatically, capping the upper tails has negatively impacted the more extreme physiological values in the real data, which are crucial at identifying severely ill patients.

A base model reproduced a similar class imbalance to the real data cohort, while an initial sampling round of 2,000 synthetic extubation episodes consisted of 127 positives, which accounted for approximately 6.35% to the extubation failure label. Furthermore, 135 synthetic episodes presented inconsistent class labels across the synthetically generated steps, which was addressed by applying majority voting on the class labels generated for the 4 steps, with the winner being copied over to each step's label that is within the same extubation episode.

A balanced synthetic dataset was then generated via rejection sampling, where the synthetic dataset consisted of 10,000 negatives and 10,000 positives. All binary variables and class labels were consistent across all steps within any one synthetic extubation episode, producing a controlled dataset without the need of increasing the weight of the minority class due to lack of coverage. The synthetic episodes belonging to the positive class have been evaluated independently because this was the class that augmentation was explored for. Considering this class alone, the feature Kolmogorov-Smirnov statistic mean resulted to be 0.156, while the normalized Wasserstein distance median was 16.691%., where 11/20 directly measured features within the positive class presented identical median with the negative class features, indicating no concrete direction of preservation. Remaining 8 features did not indicate a different change in direction between real and synthetic data, while Hemoglobin was the singular feature to suffer a reversal from +0.10 to synthetic differences registered at -0.05. Despite the results of this evaluation indicating direction preservation, complete reproduction of tail distributions was not accomplished.

An augmentation experiment consisting of 4,292 synthetic episodes with positive class label were selected from the generated samples, and added on top of the real dataset to complement the class distribution. The new training set consisted of 9,688 total extubation episodes, where the synthetic samples were combined with the real 552 episodes in the positive class, and the 4,844 negative real samples together accounted for a class balanced 50-50 training split. The validation and test sets remained untouched consisting of 1,144 and 1,666 directly measured episodes respectively. Additionally, the 20 dynamic features have been scaled using real data from the training split, while the binary features were excluded from the scaling procedure.

Even though upper winsorization presented significant overall improvement in marginal fidelity while also removing extreme outliers, the lack of tail preservation in the synthetically generated episodes indicated that these samples could not properly reproduce extreme physiological cases to represent severely ill patients, and the synthetic data was therefore utilized primarily for experimental purposes for positive class augmentation rather than a source to expand the currently class unbalanced dataset. Further downstream predictions using the augmented samples indicated that data augmentation did not improve over the baseline dataset, and was therefore not utilized for

physics-informed experiments and evaluation. A more detailed breakdown of the results produced by the models can be found in Section (6.4).

Table 6.5: Datasets Before and After Synthetic Data Augmentation using TimeDiff.

Dataset	Total Episodes	Extubation Failures	Extubation Successes	Failure Rate
Real Training Split	5,396	552	4,844	10.23%
Balanced Class Synthetic Dataset	20,000	10,000	10,000	50.00%
Synthetic Extubation Failures for Class Balancing	4,292	4,292	0	100.00%
Augmented Training Split	9,688	4,844	4,844	50.00%
Validation Split	1,144	114	1,030	9.97%
Test Split	1,166	118	1,048	10.12%

This table presents the datasets before and after augmentation with synthetic samples generated using TimeDiff.

6.4 Model Performance

The baseline models selected consisted of XGBoost, GRU, and LSTM architectures, with model evaluation being conducted using the MIMIC-IV validation split, and metrics such as AUPRC being the primary comparison mean given that samples corresponding to extubation failure only accounted for approximately 10.17% of the total MIMIC-IV cohort. Furthermore, the held-out MIMIC-IV test split was reserved for internal final evaluation of the selected models.

Various configurations were examined during the model performance evaluation stage. The physiological only features XGBoost model presented a validation AUROC score of 0.511 and an AUPRC score of 0.102, which ultimately indicate that a restricted physiological representation provided little to no predictive information for extubation failure. After applying a feature expansion step to this model, where all 20 features discussed in Section (6.2), but excluded feature source variables, scores have improved to 0.638 and 0.15 for AUROC and AUPRC respectively. Furthermore, another feature space expansion was conducted that included all 55 features, which resulted in similar results, with AUROC score of 0.637, and AUPRC of 0.151. Despite these promising results, all 3 model variations simply classified every sample as successful extubation with a default predictive threshold of 0.5, with precision, recall, and F1 scores for the extubation failure label being a flat 0.

The next experiments implemented class weighting to address the dramatic class distribution imbalance in the MIMIC-IV cohort. After this, the weighted XGBoost model presented validation results of AUROC and AUPRC of 0.646 and 0.166 respectively, which has managed to identify 73 of 114 extubation failures with the selected threshold, further achieving a recall of 0.64, precision of 0.145, F1 score of 0.236, and true negative rate of 0.582. Based on these results, we can safely claim that addressing class imbalance through a higher weighting for the minority class is necessary for the model to learn more physiologically relevant patterns, despite also producing a higher true negative rate and seeing an increase in the number of false positive predictions.

Among the 3 model architectures examined, recurrent models such as GRU and LSTM have produced the highest AUPRC scores on the internal validation set. That is, a fine tuned GRU model achieved an AUROC score of 0.643, and an AUPRC score of 0.192. Additionally, with a derived threshold cutoff of 0.458, the model also achieved precision, recall, F1, and true negative rate scores of 0.16, 0.561, 0.25, and 0.674 respectively. The fine tuned LSTM model achieved results with an AUROC score of 0.626, and AUPRC score of 0.187. Additionally, with a derived threshold cutoff of 0.464, the model also achieved higher precision and true negative rate scores of 0.19 and 0.851 respectively, but lower recall and F1 scores of 0.316 and 0.24 respectively. Based on these results, GRU has achieved the highest AUPRC score, as well as presenting higher sensitivity to extubation failures. GRU was therefore selected as the principal recurrent baseline over LSTM.

Table 6.6: Internal validation performance of baseline predictive models.

Model	AUROC	AUPRC	Precision	Recall	True Negative Rate	F1
XGBoost: Phys. Features	0.511	0.102	0.000	0.000	1.000	0.000
XGBoost: All 20 Features	0.638	0.150	0.000	0.000	1.000	0.000
XGBoost: All 55 Columns	0.637	0.151	0.000	0.000	1.000	0.000
XGBoost: All 55 Columns, Class Weighted	0.646	0.166	0.145	0.640	0.582	0.236
GRU	0.643	**0.192**	0.160	0.561	0.674	**0.249**
LSTM	0.626	0.187	**0.190**	0.316	**0.851**	0.238

*T*he base, unweighted XGBoost models were evaluated at the default threshold, which is of 0.5 and predicted NO extubation failures. GRU and LSTM models were evaluated based on derived thresholds rather than default.

As a result of evaluating the selected GRU models on the held-out MIMIC-IV test split, predictive performance decreased to an AUROC score of 0.591, and an AUPRC score of 0.136. Decision threshold was frozen at 0.458 and resulted in correctly identifying 54 of 118 extubation failures, and 700 of 1.048 successful extubations, which ultimately indicate a precision score of 0.134, recall of 0.458, F1 score of 0.21,and true negative rate of 0.668, while also producing 64 false negative, and 348 false positive predictions.

Similarly to GRU, the LSTM models also presented decreased predictive performance with an AUROC score of 0.539, and an AUPRC score of 0.106. Its decision threshold has been frozen at 0.464, and resulted in a precision score of 0.1, recall of 0.169, F1 score of 0.125, and a true negative rate of 0.827, correctly identifying 20 of 118 extubation failures and 867 of 1,048 successful extubations, as well as producing 181 false positive, and 98 false negative predictions. Based on these results, the GRU model remained the strongest performing model of the 2 when evaluated on the internal test split, despite the modest increase of an AUPRC score of 0.136 in an extubation failure distribution of approximately 10.12%. Furthermore, the results above indicate limited discrimintative capability on unseen test samples, and would therefore not consider the model as fit to be implemented in live clinical applications.

Table 6.7: Held-out MIMIC-IV Test Evaluation of GRU and LST models.

Model	Thresh.	AUROC	AUPRC	Prec.	Recall	Spec.	F1
GRU	0.458	**0.591**	**0.136**	**0.134**	**0.458**	0.668	**0.208**
LSTM	0.464	0.539	0.106	0.100	0.169	**0.827**	0.125

*T*hresholds were selected using the validation split and applied unchanged to the held-out MIMIC-IV test split. The GRU model produced 54 true positives, 700 true negatives, 348 false positives and 64 false negatives. The LSTM model produced 20 true positives, 867 true negatives, 181 false positives and 98 false negatives.

An additional exploratory sensitivity analysis was conducted on expanding the feature space with demographic features such as gender and race to see whether model predictive performance increases. After expanding the feature space, the GRU model has achieved an AUROC score of 0.587, and an AUPRC score of 0.18, compared to its no patient demographic feature counterpart with scores of 0.643 and 0.192 respectively. The LSTM model presented an AUROC score of 0.643, and an AUPRC score of 0.158, compared to its no demographic feature counterpart with scores of 0.626 and 0.187 respectively. Even though the LSTM AUROC scores presented a slight improvement, neither GRU or LSTM improved on the main AUPRC metric, and demographic features such as gender and race were therefore excluded from the final model's feature space, being retained solely for cohort description and subgroup analysis.

A data augmentation experiment was conducted using the class balanced synthetic training dataset discussed in Section (6.3) without modifying the MIMIC-IV internal validation or test splits. Training the models in a balanced class setting did not present improvements on the real

validation and test evaluations when compared to the AUPRC score of GRU when trained on no data augmentation dataset. Synthetic data augmentation was therefore excluded from any Physics-Informed experiments, not because the concept of synthetic data augmentation being ineffective, but rather because the synthetically generated extubation episodes and augmentation strategies utilized in this project did not improve the predictive performance of the underlying models.

To summarize, the fine tuned GRU model trained with no data augmentation has served as the data augmentation comparison baseline, while a fine tuned GRU model with class weight implementation will serve as the baseline for the physics-informed component implementation comparison. The physics-informed experiment that is discussed in more detail in Section (6.5) compares the specified baseline against another version of it that includes the physiological loss terms implemented within the loss logic of the model.

6.5 Physics-Informed Component Evaluation

The Physics-Informed component of the project was evaluated through a series of 6 GRU experiments, consisting of a GRU baseline model, a GRU with state prediction loss, a GRU with Alveolar gas equation loss, another GRU with oxygen content equation loss, a GRU including both equations but no state supervision, and a complete PINN model with both physiological equations. The performance of all models was evaluated on the same internal, held-out MIMIC-IV test split to ensure fairness, with thresholds being derived during validation and unchanged through testing, as well as retaining class weightings discussed in earlier sections to address the severe class imbalance present in MIMIC-IV. Furthermore, auxiliar loss components were implemented in isolation to better identify contribution of state supervision and physiological constraint.

The implemented auxiliary prediction head handled the prediction of FiO_2, $PaCO_2$, PaO_2, SpO_2, Hemoglobin, alveolar oxygen pressure, and arterial oxygen content. Training supervision availability varied significantly across the mentioned features, with SpO_2 presenting the highest availability of 98.57% withiun the available training episode steps, while FiO_2, $PaCO_2$, PaO_2, and Hemoglobin presented availabilities of 43.19%, 12.08%, 12.08%, and 5.59% respectively. Therefore, due to a lack in availability across these features, the targets derived by the alveolar oxygen pressure values could be reconstructed for approximately 5.85%, and roughly 2.25% for the arterial oxygen content values. Masking was implemented to ensure state supervision losses are computed only when respective targets are available. Lastly, the configuration consisting of the 2 physiological constraints was selected based on the results presented after conducting validation experiments with the 2 physiological equations. The state supervision weight was set to $\alpha_{\text{state}} = 0.01$, the alveolar gas equation weight to $\beta_{\text{alveolar}} = 0.05$, and the oxygen content equation weight to $\beta_{\text{oxygen}} = 0.01$, with the best validation performance of this configuration being identified at the 28th epoch.

The experiments resulted in a slight variation in the validation AUPRC score across the 6 models, with the lowest score being achieved by the GRU model with Oxygen Content Equation Loss, and the highest one of 0.1946 achieved by the GRU model with both equations loss, but no anchoring, while the baseline GRU model has achieved an AUROC score of 0.661 and an AUPRC score of 0.1924. The implementation of state supervision presented an improvement in the AUPRC score reaching 0.1945, while the GRU models constrained by the Alveolar Gas Equation and the Oxygen Content Equation independently have obtained AUPRC scores of 0.1928 and 0.1909 respectively. Lastly, the full PINN model has achieved a validation AUROC score of 0.6589, and an AUPRC score of 0.1938. Therefore, based on these results, none of the model configurations has produced a substantial improvement.

Table 6.8: Validation performance and selected operating thresholds for the physics informed ablation of GRU models.

Model	Threshold	Validation AUROC	Validation AUPRC
Baseline GRU	0.6707	0.6610	0.1924
GRU with State Predictions Loss	0.5891	0.6608	0.1945
GRU with Alveolar Gas Equation Loss	0.5226	0.6534	0.1928
GRU with Oxygen Content Equation Loss	0.5546	0.6577	0.1909
GRU with Both Equations Loss, No Anchoring	0.5827	0.6552	0.1946
Full PINN	0.5892	0.6589	0.1938

*T*his table presents internal validation conducted on MIMIC-IV data of the various physics-informed implementations attached to the GRU model.

The evaluation conducted on the held-out MIMIC-IV test split exhibited similar patterns with the validation results. That is, the baseline GRU model has achieved an AUROC score of 0.6017, and an AUPRC score of 0.1357, and given the decision threshold of 0.6707 derived on the validation split, the model achieved precision, recall, F1, and true negative rate scores of 0.1469, 0.1780, 0.161, and 0.8836 respectively, as well as indicating a clear increase of the recall score to 0.3644, and F1 score to 0.2038 through state supervision, while the true negative rate has decreased to 0.751.

The GRU model constrained by the Alveolar Gas Equation has achieved the highest test performance with an AUROC score of 0.6119, and AUPRC score of 0.1367, as well as a recall of 0.3390 and an F1 score of 0.2046. Not far behind, the Oxygen Content Equation constrained GRU model has achieved a recall score of 0.3305 and an F1 score of 0.2, but also exhibited a decrease in AUPRC score to 0.1311. Furthermore, the unanchored GRU model constrained by both physiological equations has achieved an AUROC score of 0.6086 and AUPRC of 0.1335, but limited improvements in terms of recall and F1 score, as presented in Table (6.9).

Lastly, the full PINN model has achieved an AUROC score of 0.6021, and an AUPRC score of 0.1350, which indicates the exact scores of the baseline GRU model, but with a significant increase in recall from 0.1780 to 0.3729, as well as in F1 score from 0.1609 to 0.2075. Additionally, the model presented a decrease in the true negative rate from 0.8836 to 0.75, as well as in precision from 0.1469 to 0.1438. The full PINN model has managed to correctly identify 44 of 118 extubation failure labels, and 786 of 1,048 successful extubation ones, as well as predicting 262 false positives and 74 false negatives.

Table 6.9: Performance evaluation on Held-out MIMIC-IV test split of physics informed ablation of GRU models.

Model	Precision	Recall	True Negative Rate	F1	AUROC	AUPRC
Baseline GRU	0.1469	0.1780	0.8836	0.1609	0.6017	0.1357
GRU with State Predictions Loss	0.1414	0.3644	0.7510	0.2038	0.6032	0.1354
GRU with Alveolar Gas Equation Loss	0.1465	0.3390	0.7777	0.2046	0.6119	0.1367
GRU with Oxygen Content Equation Loss	0.1434	0.3305	0.7777	0.2000	0.6029	0.1311
GRU with Both Equations Loss, No Anchoring	0.1413	0.2203	0.8492	0.1722	0.6086	0.1335
Full PINN	0.1438	0.3729	0.7500	0.2075	0.6021	0.1350

*T*his table presents the evaluation metrics resulted by the physics informed ablation of GRU models on the internal held out test split, where each model was evaluated using its validation-derived threshold.

Further analysis aimed to estimate uncertainty in differences between the complete PINN model and the baseline GRU was conducted using 2,000 bootstrapped resamples based on the internal held-out MIMIC-IV test set. Patient sampling was done using replacement, where all extubation episodes belonging to the same patient were retained together. To evaluate the 2 models, each model was evaluated on each resample, while the resulting distribution differences were used to compute the 95% confidence analysis. The results presented an increase in recall score to 0.1949, with a 95% confidence interval ranging between 0.1262 and 0.2672. Additionally, there was a -0.1336 decrease in the true negative rate, with the confidence interval ranging between -0.1538 and -0.1134. In contrast, confidence intervals for scores such as AUROC, AUPRC, precision, amd F1 included 0 within the interval, therefore not indicating support in the observed change of the scores. More specifically, the F1 confidence interval ranged from -0.0034 to 0.0966, accounting for a total difference of 0.0466, while total differences in AUROC and AUPRC were 0.0004, and -0.0007 respectively. In conclusion, although the model presented a significant increase in the recall score, it should be interpreted as a change in a threshold determined configuration rather than pure evidence of improved discriminative capability.

Analysis of the equation residuals presented a more significant impact of the physiological constraints, where the baseline GRU displayed an alveolar gas equation residual mean absolute error of 33.2664 mmHg, and an oxygen content equation resual mean absolute error of 0.3595 mLO_2/dL. Additionally, state supervision has managed to further reduce the alveolar gas residual to 15.4874 mmHg, but no further decreases resulted for the oxygen content residual.

Essentially, each individual physiological constraint has managed to improved consistency within its corresponding physiological equation, with the alveolar gas equation constraint model presenting a reduction in its alveolar gas residual to 0.1243 mmHg, while also maintaining an oxygen content residual of 0.3662 mLO_2/dL. In contrast, the oxygen content equation constraint model has also managed to reduce its corresponding oxygen content residual to 0.0056 mLO_2/dL, while maintaining an alveolar gas residual of 17.0273 mmHg.

The full PINN model has managed to reduce both residuals simmultaneously, resulting in an alveolar gas residual of 0.8805 mmHg, and an oxygen content residual of 0.0518 mLO_2/dL, which in terms of state supervision, the model presented reductions of approximately 94.3% and 87.0% respectively. While the unanchored model constrained by both equations has achieved even lower residuals of 0.1249 mmHg and 0.0009 mLO_2/dL, a lack in state supervision indicates that a guaranteed correspondence between the predictions and the clinical measurements cannot be established because a low residual can not determine that the predicted states have consistently remained representative with the real patient data. In conclusion, the unanchored model was retained for comparison purposes.

Table 6.10: Mean absolute errors of the equation residuals in GRU physics informed ablations evaluated on the internal held-out MIMIC-IV test split.

Model	**Alveolar residual MAE (mmHg)**	**Oxygen-content residual MAE (mL O_2/dL)**
Baseline GRU	33.2664	0.3595
GRU with State Predictions Loss	15.4874	0.3978
GRU with Alveolar Gas Equation Loss	0.1243	0.3662
GRU with Oxygen Content Equation Loss	17.0273	0.0056
GRU with Both Equations Loss, No Anchoring	0.1249	0.0009
Full PINN	0.8805	0.0518

*T*his table presents the mean absolute errors of the equation residuals in GRU physics informed ablation evaltions. Ideally, these values would be on the lower end to indicate consistency with the underlying physiological equation.

Based on the results discussed within this section, we can claim that the 2 physiological constraints have successfully improved consistency with the 2 physiological equations, but according to a lack of improvement in AUROC and AUPRC scores, it did not actively improve predictive capability. The most significant impact on the discriminative capability of the models was to identify

a larger amount of extubation failure labels at the cost of more false positive predictions. A more detailed discussion on the trade-offs and impact of the physiological constraints on the predictive performance of the models are presented in Section (7.3).

6.6 External Validation

The external validation cohort derived from eICU-CRD consisted of a total of 13,595 valid extubation episodes, split into 909 extubation failures and 12,686 successful extubations. The resulting positive class distribution was of approximately 6.69%, which is below MIMIC-IV's 10.12% held-out test split. The chosen full PINN model was applied without modifying its MIMIC-IV derived weights, or recalibrating its MIMIC-IV decision threshold of 0.5892.

As a result of this analysis, the model achieved an external AUROC score of 0.5515, and an AUPRC score of 0.0792, which was only slightly above the positive class distribution of 0.0669 within the dataset. Compared to the model's performance on the MIMIC-IV held-out test split, the AUROC score has decreased from 0.6021 down to 0.5515, and the AUPRC from 0.1350 to 0.0792, therefore indicating a clear weaker ability to discriminate between class labels on the external validation cohort.

Table 6.11: External validation performance of the selected two physiological constraint physics-informed GRU on the eICU-CRD dataset.

Metric	Estimate	95% Confidence Interval
AUROC	0.5515	0.5305-0.5709
AUPRC	0.0792	0.0726-0.0880
Precision	0.0924	0.0692-0.1172
Recall	0.0561	0.0417-0.0716
Specificity	0.9605	0.9570-0.9640
F1-score	0.0698	0.0521-0.0883

*T*his table presents the evaluation results of external validation performance of the selected two physiological constraint physics-informed GRU on the eICU-CRD dataset. Confidence intervals were calculated using 2,000 episode-level bootstrap resamples.

The model has managed to correctly identify 51 of 909 extubation failure labels, which resulted in a recall score of 0.0561. On the other hand, it has correctly classified 12,185 of 12,686 successful extubations, resulting in a true negative rate of 0.9605. Additionally, 501 successful extubations were wrongly classified as failures, and 858 extubation failures were wrongly classified as successful extubations, therefore producing a precision score of 0.0924, and an F1 score of 0.0698.

Compared to the results obtained on the held-out MIMIC-IV test split, recall has decreased from 0.3729 to 0.0561, precision dropped from 0.1438 to 0.0924, and F1 score also decreased from 0.2075 to 0.0698. In contrast, the true negative rate has improved from 0.75 to 0.9605 since the model has classified fewer eICU-CRD extubation episodes as extubation failures. Overall, the model presented a reduction in predictive performance with a particularly low recall score indicate limited transportability of the model to eICU-CRD.

Chapter 7

Discussion

7.1 Predictive Performance and External Transportability

Based on the baseline experiments, results indicated that class imbalance was a strong factor in the negative impacts affecting the models' predictive capabilities. Expanding the XGBoost features space resulted in improved AUROC and AUPRC scores, therefore indicating that variables outside of the alveolar gas and arterial oxygen content equations contributed positively. However, while using the default decision threshold of 0.5, all unweighted XGBoost configurations simply classified every extubation episode as a successful extubation, indicating the need for class balancing. Class weighting was incorporated to address the issue, resulting in the model being able to predict extubation failures, but this increase in recall score also resulted in a decrease in precision a substantial amount of false positive predictions, indicating that predictive and threshold performance should be considered together when evaluating an imbalanced dataset.

Recurrent model architectures such as GRU and LSTM have produced the highest validation AUPRC scores, with GRU outperforming LSTM and the class weigthed XGBoost models. GRU's AUPRC score has decreased from 0.192 on the validation split to 0.136 on the held-out MIMIC-IV internal evaluation test split, while AUROC has also decreased from 0.643 to 0.591. Therefore, the test AUPRC score only scored sightly above the extubation failure class distribution of 0.1012, suggesting that the 4 time step ventilation episode representation consisted of predictive signals, but insufficient to capture all details to determine patient extubation outcome. Other factors such as aiway protection, secretion burden, and cough strength to name a few, were not completely represented in the feature space, and compared with previous studies that achieved stronger performance, this study is limited by the 6 hour decision window, feature space, as well as extubation outcome definitions.

External validation required more generalizability from the developed model. The selected physics-informed GRU model with 2 physiological constraints has achieved an AUROC score of 0.6021, and an AUPRC score of 0.135 on the MIMIC-IV held-out test split, but these decreased to 0.5515 and to 0.0792 respectively on the eICU-CRD cohort. Given the MIMIC-IV derived decision threshold, recall has also decreased from 0.3729 to 0.0561, correctly identifying 51 of 909 external extubation failures. The true negative rate has increased from 0.75 to 0.9605 because eICU-CRD contains fewer extubation failure episodes than MIMIC-IV. The lower distribution of the positive class identified in eICU-CRD clearly explains the reduction in AUPRC and precision scores.

Other factors such as differences in the patient population, clinical practice, data collection procedures, hospital systems, and measurement frequency also play an important role in decrease in predictive performance, and they could not be individually isolated and addressed or explored within the current evaluation protocols. Overall, external validation results indicate that there is limited transportability across distict datasets, and that performance within a dataset, in this case MIMIC-IV, can not be assumed to be consistently generalizable when applied to a distinct ICU dataset. That being said, the model should be treated as an exploratory experiment investigating whether synthetic data augmentation for class balancing, and physiological constraints can improve predictive performance.

7.2 Impact of Synthetic Data Augmentation

The initial TimeDiff model properly reproduced the median regions of several real feature distributions but generated a few dramatic upper bound extremes for some of the ventilator settings features. Applying a winzorization of the upper tail based on training samples has significantly reduced the extreme synthetic values, also presenting a decrease in the mean Kolmogorov-Smirnov statistic from 0.194 tp 0.156, and diminishing the difference between synthetic median and 99th percentile values and their corresponding real measurements. These results indicate that even a small amount of extreme real measurements could have a negative impact on the diffusion generative model performance, and that upper tail capping applied during the preprocessing stage can improve fidelity of the synthetically generated samples.

Despite these improvements in marginal distribution similarity, it does not automatically mean that a complete clinical distribution has been successfully reproduced. Several synthetically generated upper bounds suffered of compression, for example features such as FiO_2, PEEP, heart rate, and ventilation duration, which is important because clinically plausible extreme values may represent severely ill patients that the generative model attempted to capture. Furthermore, inconsistency in the synthetically generated outcome labels across the episode time steps as a result of the unconditional TimeDiff model suggests that the model did not properly learn the time step per episode structure of the outcome label. Lastly, preservation of 8 out of 9 class differences with a non zero median has only provided limited evidence of class fidelity, given a simplistic nature and not assessing more complex relationships between the temporal trajectory of feature values.

Applying balanced rejection sampling to address class imbalance within the training split has managed to provide the ability to generate perfectly balanced datasets, but did not ultimately mean that TimeDiff the synthetically generated extubation failure samples properly represented clinically relevant minority class cases. By adding approximately 4,292 synthetically generated extubation failure episodes did not improve upon the directly measured MIMIC-IV validation set, which peaked with an AUPRC score of 0.192. Results of this analysis indicate that a class imbalance with a limited number extubation failures was not the sole limitation of the model's predictive performance. Additionally, this indicates that while the synthtetically generated data may have increased the number of extubation failure labels, the added samples did not necessarily consist of sufficiently accurate clinical information to support the physiological patterns and improve discrimination between the negative and positive class.

Overall, these findings do not establish that synthetic data augmentation is ineffective for patient extubation readiness prediction, but rather indicate this study's configuration consisting of a specific preprocessing strategy, and rejection sampling to address class imbalance did not improve predictive performance on the real MIMIC-IV validation split. Therefore, synthetic data augmentation was not further conducted during the physics-informed experiments, but also because it would have been more difficult to distinguish between the effects of physiological constraints and the synthetic data augmentation during experiments.

7.3 Impact of Physiological Constraints

The validation results of the physics-informed ablation indicated that by incorporating physiological supervision alone, little to no change occurs in the predictive performance. The validation AUPRC score varied very modestly between 0.1909 and 0.1946 across all 6 model configurations, with the baseline model achieving 0.1924, and the full PINN model achieving 0.1938. That being said, simple physiological losses did not provide a strong advantage in discriminating between extubation failures and seccessful extubations.

After evaluating on the held-out test split, the primary change resulted in the full PINN model was in the threshold dependent metrics, which compared to the baseline GRU model, the recall score has increased from 0.1780 to 0.3729, and F1 score increased from 0.1609 to 0.2075, while precision remained the same, and the true negative rate has decreased from 0.8836 to 0.75. Furthermore, the paired bootstrap analysis exhibited clear support of the recall score increase as well as decrease in the true negative rate because the associated confidence intervals excluded 0, while

the confidence intervals of metrics such as F1 score, AUROC, and AUPRC did include 0. As a result, the model indicated more sensitivity to extubation failures, but no concrete evidence exists to safely claim that the physiological constraints improved the overall predictive performance. Lastly, the physics-informed constrained was planned to be based on a mathematical pulmonary gas exchange model developed by researchers at Aarhus University, in Denmark, who have partnered with Imperial College London, but due to the time constraints of this project, the physics-informed component was limited to the alveolar gas and arterial oxygen content equations.

Residual analysis results has confirmed that the implemented equation based constraints directly influenced the physiological branch as planned. Compare to the baseline of only state supervision, the full PINN model has managed to reduce the alveolar gas equation residual by approximately 94.3%, as well as the arterial oxygen content equation residual by 87%, where each singular equation model configuration primarily reduced the corresponding physiological equation residual. The model unanchored by state supervision has achieved even smaller quation residuals, but the resulting physiological states were not evaluated against real measurements. Therefore, demonstrating that low physiological equation residuals do not automatically provide sufficient evidence that predicted physiological states are clinically accurate.

A potential explanation of the limited improvement in the predictive capability of the underlying models is that the equations only describe the pulmonary gas exchange component of the extubation readiness problem. Based on past works, extubation failure may also be dependent on other factors such as airway protection, secretion management, neurological status, and treatment to name a few. Additionally, $PaCO_2$, PaO_2, and hemoglobin were amongst the least available features and heavily depended on filling and imputation, further limiting the physiological branch's exposure to real measurements. In conclusion, while the equation based constraints significantly improved consistency with the underlying physiological equations, there was no impact on the discriminative capability of the model, suggesting that physiological consistency should be treated as a model property rather than evidence of improvement in the model's predictive performance.

7.4 Limitations

This study was conducted on retrospective intensive care records collected routinely during clinical practice rather than specifically for presenting a research question. Ventilation episodes, patient extubation outcomes, and post extubation respiratory both invasive and non-invasive had to be reconstructed from the labels, variables, and timestamps available within each dataset. The extubation failure definition consisted of both reintubation and prolonged non-invasive respiratory support within and for longer than 48 hours after the extubation event respectively, which can be easily impacted by different clinical mechanisms, and differences in local clinical treatment practice. Additionally, splitting the extracted extubation episodes on a patient basis to prevent leakage in the MIMIC-IV cohort was successful primarily because the data originated from the same hospital system, while deterioration could be observed when the extracted relationships were not as consistent within eICU-CRD, which indicates poor transferability across clinical environments.

Another substantial limitation was the lack of feature coverage in the defined decision window of 6 hours prior to the extubation event, where $PaCO_2$, PaO_2, and hemoglobin were poorly available and heavily depended on MICE imputations. While incorporating the value source binary variables helped the model distinguish between where values originated, they do not address high dependency on imputation, which was crucial for the physiological branch because the measured values had a direct impact on the state supervision and equation based loss terms. Furthermore, the implemented physiological equations represented simplified relationships, where terms such as atmospheric pressure, water vapor pressure, and respiratory quotient were declared fixed constants, and SpO_2 was treated as an available approximation of the arterial oxygen saturation. Lastly, as discussed in previous sections, low residual values only indicate agreement with the underlying physiological equation rather than clinical accuracy.

The chosen structure of 4 time step representations per episode provided a compressed version of the 6 hour decision window prior to the extubation event, which could have easily missed long term trends or short term spikes that could hold clinical information. The data augmentation

experiments analyzed the marginal distributions and median relationships across features, but an improved Kolmogorov-Smirnov statistic and Wasserstein Distance do not that guarantee that the generated samples cover all extreme value possibilities, which may indicate the severely ill patients. Furthermore, confidence intervals were computed using bootstrap resampling at the episode level, which did not account for multiple episodes that may belong to the same underlying patient.

Lastly, no formal causal or counterfactual analysis was conducted in the study. While input variable alteration could be performed to observe how the prediction changes, and ultimately identify the model's sensitivity to that specific variable, it would not be able to provide a clinical explanation of the effects caused by changing a ventilator setting. Other variables, such as FiO_2, PEEP and pressure support level could create a potential confounding because they are influenced by both patient state of severity as well as clinical decisions. Incorporating the physiological equations does not automatically remove the confounding nor establishes causility, and the model's predictive output should be treated as a physiologically grounded estimate rather than a clinical decision.

7.5 Clinical and Methodological Implications

As mentioned previously, the predictive performance achieved throughout this study should not be treated as an independent clinical decision. While the full PINN model has managed to identify more extubation failures than the comparable baseline on the MIMIC-IV held-out test split, the precision score of 0.1438 means that several predictions of the positive class were false positives. In clinical practice, if an extubation failure prediction is missed, there is a high risk that the patient is subjected to reintubation and the associated complications of the procedure unnecessarily. Additionally, a recall score of 0.0561 is rather low to be considered a reliable screenning tool.

While the conducted experiments resulted in physiological consistency with the underlying equations, predictive capability should be treated as a separate component, because the implemented constraints reduced the equation residuals significantly, and balanced the recall and true negative rate relationship, there was no reliable improvement in the AUROC and AUPRC scores. Additionally, the unanchored model has achieved even smaller residuals, but excluded the supervision state and therefore indicated that simply satisfying a physiological relation does not automatically guarantee predicted physiological states are clinically meaningful. Implementing the full complexity of the physiological branch is not justified if the sole purpose of the study is to maximize the model's predictive performance, but it provides a physiologically grounded representation of pulmonary gas exchange relationships.

Synthetic data augmentation and the external validation experiments provide crucial methodological findings, such as that there is no guarantee of synthetic samples will improve the predictive capability of the model on real samples if marginal similarity and class balance are improved. Additionally, validation of a model's predictive performance on a single dataset is notsufficient evidence of generalizability. We also learn that differences in class coverage, measurement practices, and feature distribution significantly impacted the model's performance when tested against eICU-CRD.

Chapter 8

Conclusions and Future Work

8.1 Conclusions

This project explored whether synthetic data augmentation and physiological constraints grounded by pulmonary gas exchange equations can improve prediction capabilities of patient extubation failure from routinely collected intensive care records. An end to end workflow has been developed consisting of data suitability assessment, valid cohort construction, preprocessing time-series data, baseline predictive modeling, synthetic EHR data generation, physics-informed led model learning, and external validation. MIMIC-IV was the primary dataset used for model development and internal validation, while eICU-CRD was used for external validation aiming to assess model transportability.

Baseline model experiments indicated that an expanded feature space, and explicitly addressing class imbalancing were crucial factors in improving a model's predictive capability of extubation failures. Recurrent models such as GRU and LSTM have achieved the highest validation performance, while predictive performance on on the held-out internal dataset remained modest. TimeDiff was the generative model of choice, which was used to generate a large pool of synthetic ventilation episodes, where marginal statistical fidelity has been improved through upper winsorization. Furthermore, incorporating synthetic samples into training did not improve the predictive performance on the real validation split, suggesting that simply increasing the number of samples in the minority class is not sufficient if the synthetic samples do not fully preserve the relevant clinical and time-series information.

The incorporated physiological constraints have significantly reduced violations of the alveolar gas and arterial oxygen content equations. When compared to the baseline GRU, the full PINN model exhibited an increase in the recall score from 0.178 to 0.3729, but also a decrease in the true negative rate from 0.8836 to 0.75, with no significant improvement in the AUROC and AUPRC scores. As a result of the external validation analysis conducted using the eICU-CRD dataset, its predictive performance decreased to an AUROC score of 0.5515, an AUPRC score of 0.0792, and a recall of 0.0561. In conclusion, the project successfully demonstrates that physiological consistency can alter the model's behavior without improving the predictive performance, as well as that performance results gathered from one dataset does not necessarily guarantee high generalizability. Therefore, this project provides a controlled experimental workflow that explores the methods described above, but it does not currently support reliable independent clinical application.

8.2 Future Work

Future research on this matter is suggested to explore whether richer remporal and clinical information can improve predictive performance. The current 4 time step per episode structure can be expanded by potentially using longer observation periods, while also expanding the feature space to include variables such as spontaneous breathing trial results, cough strength, secretion burden, and neurological status. Additionally, evaluation could furhter employ additional datasets such as Hirid, or AmsterdamUMCdb if time allows. Lastly, other metrics such as decision curve analysis and patient cluster uncertainty estimations may provide more insightful information on

the model's behavior.

Synthetic data augmentation can be revisited using a class conditional generative model that is trained specifically to model extubation failures, rather than applying rejection sampling. Extreme value methods and weather forecasting approaches could be explored to help build a better representation of severely ill patients, which were essentially compressed in TimeDiff. Evaluation metrics could also pe expanded beyond marginal distribution to other ones such as temporal dependency, and class conditional fidelity analysis.

Improvements to the physics-informed component can be potentially added by explicitly accounting for measurement uncertainty. That is, the equation and state supervision loss terms can provide better confidence for real samples rather than filled or imputed values. Additionally, if available, arterial oxygen saturation can be used instead of SpO_2, as well as investigating other constraints representing respiratory mechanics, such as acid-base physiology. However, these implementations should be carefully examined, and isolating them to determine the amount of contribution each brings to the predictive performance.

Lastly, patient extubation management could be defined as a temporal causal problem, where strucutral causal models could be used to represent ventilator settings changes, patient severity state, as well as time bounded confounding. This formulation would provide a more appropriate baseline for a counterfactual analysis, but noting that any counterfactual recommendation would still need to be clinically assessed, and properly validated.

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